

Chapter 10

BOILING AND CONDENSATION

Boiling Heat Transfer

10-1C Boiling is the liquid-to-vapor phase change process that occurs at a solid-liquid interface when the surface is heated above the saturation temperature of the liquid. The formation and rise of the bubbles and the liquid entrainment coupled with the large amount of heat absorbed during liquid-vapor phase change at essentially constant temperature are responsible for the very high heat transfer coefficients associated with nucleate boiling.

10-2C Yes. Otherwise we can create energy by alternately vaporizing and condensing a substance.

10-3C Both boiling and evaporation are liquid-to-vapor phase change processes, but evaporation occurs at the *liquid-vapor interface* when the vapor pressure is less than the saturation pressure of the liquid at a given temperature, and it involves no bubble formation or bubble motion. Boiling, on the other hand, occurs at the *solid-liquid interface* when a liquid is brought into contact with a surface maintained at a temperature T_s sufficiently above the saturation temperature T_{sat} of the liquid.

10-4C Boiling is called *pool boiling* in the absence of bulk fluid flow, and *flow boiling* (or *forced convection boiling*) in the presence of it. In pool boiling, the fluid is stationary, and any motion of the fluid is due to natural convection currents and the motion of the bubbles due to the influence of buoyancy.

10-5C Boiling is said to be *subcooled* (or *local*) when the bulk of the liquid is subcooled (i.e., the temperature of the main body of the liquid is below the saturation temperature T_{sat}), and *saturated* (or *bulk*) when the bulk of the liquid is saturated (i.e., the temperature of all the liquid is equal to T_{sat}).

10-6C The boiling curve is given in Figure 10-6 in the text. In the *natural convection boiling* regime, the fluid motion is governed by natural convection currents, and heat transfer from the heating surface to the fluid is by natural convection. In the *nucleate boiling* regime, bubbles form at various preferential sites on the heating surface, and rise to the top. In the *transition boiling* regime, part of the surface is covered by a vapor film. In the *film boiling* regime, the heater surface is completely covered by a continuous stable vapor film, and heat transfer is by combined convection and radiation.

10-7C In the *film boiling* regime, the heater surface is completely covered by a continuous stable vapor film, and heat transfer is by combined convection and radiation. In the nucleate boiling regime, the heater surface is covered by the liquid. The boiling heat flux in the stable film boiling regime can be higher or lower than that in the nucleate boiling regime, as can be seen from the boiling curve.

10-8C The boiling curve is given in Figure 10-6 in the text. The burnout point in the curve is point C. The *burnout* during boiling is caused by the heater surface being blanketed by a continuous layer of vapor film at increased heat fluxes, and the resulting rise in heater surface temperature in order to maintain the same heat transfer rate across a low-conducting vapor film. Any attempt to increase the heat flux beyond $\dot{q}_{\max}$ will cause the operation point on the boiling curve to jump suddenly from point C to point E. However, the surface temperature that corresponds to point E is beyond the melting point of most heater materials, and burnout occurs. The burnout point is avoided in the design of boilers in order to avoid the disastrous explosions of the boilers.

10-9C Pool boiling heat transfer can be increased *permanently* by increasing the number of nucleation sites on the heater surface by *coating* the surface with a thin layer (much less than 1 mm) of very porous material, or by *forming cavities* on the surface mechanically to facilitate continuous vapor formation. Such surfaces are reported to enhance heat transfer in the nucleate boiling regime by a factor of up to 10, and the critical heat flux by a factor of 3. The use of finned surfaces is also known to enhance nucleate boiling heat transfer and the critical heat flux.

10-10C The different boiling regimes that occur in a vertical tube during flow boiling are forced convection of liquid, bubbly flow, slug flow, annular flow, transition flow, mist flow, and forced convection of vapor.

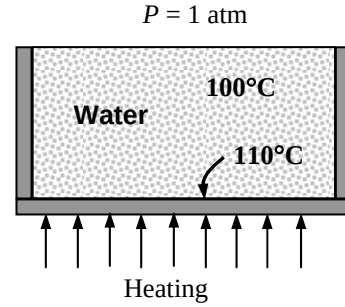
10-11 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a mechanically polished stainless steel pan whose inner surface temperature is maintained at $T_s = 110^\circ\text{C}$. The rate of heat transfer to the water and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 110 - 100 = 10^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(110 - 100)}{0.0130(2257 \times 10^3)^{1.75}} \right)^3 \\ &= 140,700 \text{ W/m}^2\end{aligned}$$

The surface area of the bottom of the pan is

$$A_s = \pi D^2 / 4 = \pi (0.25 \text{ m})^2 / 4 = 0.04909 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.04909 \text{ m}^2)(140,700 \text{ W/m}^2) = \mathbf{6907 \text{ W}}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{6907 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{3.06 \times 10^{-3} \text{ kg/s}}$$

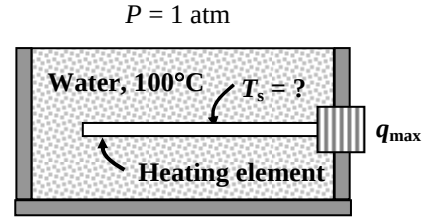
That is, water in the pan will boil at a rate of 3 grams per second.

10-12 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a mechanically polished stainless steel heating element. The maximum heat flux in the nucleate boiling regime and the surface temperature of the heater for that case are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eqs. 10-2 and 10-3 in connection with their definitions in order to avoid unit manipulations. For a large horizontal heating element, $C_{cr} = 0.12$ (Table 10-4). (It can be shown that $L^* = 5.99 > 1.2$ and thus the restriction in Table 10-4 is satisfied).

Analysis The maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12 (2257 \times 10^3) [0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= \mathbf{1,017,000 \text{ W/m}^2}\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into the Rohsenow relation together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,017,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = \mathbf{119.3^\circ\text{C}}$$

Therefore, the temperature of the heater surface will be only 19.3°C above the boiling temperature of water when burnout occurs.

10-13 "PROBLEM 10-13"**"GIVEN"**

D=0.003 "[m]"

"P_{sat}=101.3 [kPa], parameter to be varied"**"PROPERTIES"**

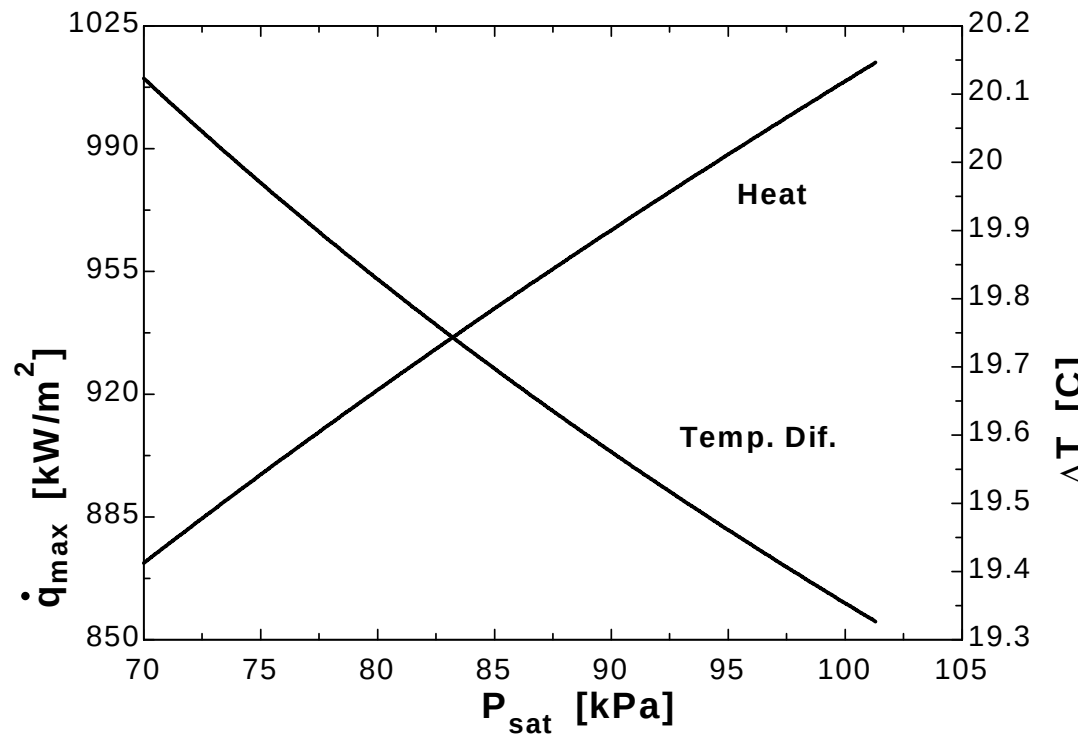
Fluid\$='steam_NBS'

T_{sat}=temperature(Fluid\$, P=P_{sat}, x=1)rho_l=density(Fluid\$, T=T_{sat}, x=0)rho_v=density(Fluid\$, T=T_{sat}, x=1)sigma=SurfaceTension(Fluid\$, T=T_{sat})mu_l=Viscosity(Fluid\$, T=T_{sat}, x=0)Pr_l=Prandtl(Fluid\$, T=T_{sat}, P=P_{sat})C_l=CP(Fluid\$, T=T_{sat}, x=0)h_f=enthalpy(Fluid\$, T=T_{sat}, x=0)h_g=enthalpy(Fluid\$, T=T_{sat}, x=1)h_{fg}=h_g-h_fC_{sf}=0.0130 "from Table 8-3 of the text"

n=1 "from Table 8-3 of the text"

C_{cr}=0.12 "from Table 8-4 of the text"g=9.8 "[m/s²], gravitational acceleraton"**"ANALYSIS"**q_{dot_max}=C_{cr}*h_{fg}*(sigma*g*rho_v²*(rho_l-rho_v))^{0.25}q_{dot_nucleate}=q_{dot_max}q_{dot_nucleate}=mu_l*h_{fg}*(((g*(rho_l-rho_v))/sigma)^{0.5})*((C_l*(T_s-T_{sat}))/(C_{sf}*h_{fg}*Pr_lⁿ))³DELTA T=T_s-T_{sat}

P _{sat} [kPa]	q _{max} [kW/m ²]	ΔT [C]
70	871.9	20.12
71.65	880.3	20.07
73.29	888.6	20.02
74.94	896.8	19.97
76.59	904.9	19.92
78.24	912.8	19.88
79.88	920.7	19.83
81.53	928.4	19.79
83.18	936.1	19.74
84.83	943.6	19.7
86.47	951.1	19.66
88.12	958.5	19.62
89.77	965.8	19.58
91.42	973	19.54
93.06	980.1	19.5
94.71	987.2	19.47
96.36	994.1	19.43
98.01	1001	19.4
99.65	1008	19.36
101.3	1015	19.33

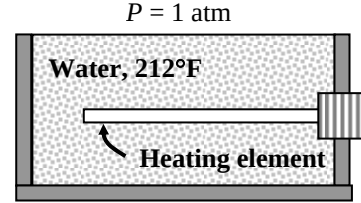


10-14E Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 212^\circ\text{F}$ by a horizontal polished copper heating element whose surface temperature is maintained at $T_s = 788^\circ\text{F}$. The rate of heat transfer to the water per unit length of the heater is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 212°F are $\rho_l = 59.82 \text{ lbm/ft}^3$ and $h_{fg} = 970 \text{ Btu/lbm}$ (Table A-9E). The properties of the vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (212 + 788)/2 = 500^\circ\text{F}$ are (Table A-16E)

$$\begin{aligned}\rho_v &= 0.02571 \text{ lbm/ft}^3 \\ \mu_v &= 0.04564 \text{ Btu/lbm} \cdot \text{h} \\ C_{pv} &= 0.4707 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_v &= 0.02267 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Also, $g = 32.2 \text{ ft/s}^2 = 32.2 \times (3600)^2 \text{ ft/h}^2$. Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations. Also note that we used vapor properties at 1 atm pressure from Table A-16E instead of the properties of saturated vapor from Table A-9E since the latter are at the saturation pressure of 680 psia (46 atm).

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 788 - 212 = 576^\circ\text{F}$, which is much larger than 30°C or 54°F . Therefore, film boiling will occur. The film boiling heat flux in this case can be determined to be

$$\begin{aligned}\dot{q}_{\text{film}} &= 0.62 \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 C_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{32.2 (3600)^2 (0.02267)^3 (0.02571) (59.82 - 0.02571) [970 + 0.4 \times 0.4707 (788 - 212)]}{(0.04564) (0.5/12) (788 - 212)} \right]^{1/4} \times (788 - 212) \\ &= 18,600 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

The radiation heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.08) (0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4) [(788 + 460 \text{ R})^4 - (212 + 460 \text{ R})^4] \\ &= 305 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

Note that heat transfer by radiation is very small in this case because of the low emissivity of the surface and the relatively low surface temperature of the heating element. Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 18,600 + \frac{3}{4} \times 305 = 18,829 \text{ Btu/h} \cdot \text{ft}^2$$

Finally, the rate of heat transfer from the heating element to the water is determined by multiplying the heat flux by the heat transfer surface area,

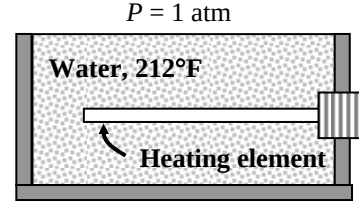
$$\begin{aligned}\dot{Q}_{\text{total}} &= A_s \dot{q}_{\text{total}} = (\pi D L) \dot{q}_{\text{total}} \\ &= (\pi \times 0.5/12 \text{ ft} \times 1 \text{ ft}) (18,829 \text{ Btu/h} \cdot \text{ft}^2) \\ &= \mathbf{2465 \text{ Btu/h}}\end{aligned}$$

10-15E Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 212^\circ\text{F}$ by a horizontal polished copper heating element whose surface temperature is maintained at $T_s = 988^\circ\text{F}$. The rate of heat transfer to the water per unit length of the heater is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 212°F are $\rho_l = 59.82 \text{ lbm/ft}^3$ and $h_{fg} = 970 \text{ Btu/lbm}$ (Table A-9E). The properties of the vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (212 + 988)/2 = 600^\circ\text{F}$ are, by interpolation, (Table A-16E)

$$\begin{aligned}\rho_v &= 0.02395 \text{ lbm/ft}^3 \\ \mu_v &= 0.05101 \text{ Btu/lbm} \cdot \text{h} \\ C_{pv} &= 0.4799 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_v &= 0.02640 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Also, $g = 32.2 \text{ ft/s}^2 = 32.2 \times (3600)^2 \text{ ft/h}^2$. Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations. Also note that we used vapor properties at 1 atm pressure from Table A-16E instead of the properties of saturated vapor from Table A-9E since the latter are at the saturation pressure of 1541 psia (105 atm).

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 988 - 212 = 776^\circ\text{F}$, which is much larger than 30°C or 54°F . Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= 0.62 \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 C_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{32.2 (3600)^2 (0.02640)^3 (0.02395) (59.82 - 0.02395) [970 + 0.4 \times 0.4799 (988 - 212)]}{(0.05101) (0.5/12) (988 - 212)} \right]^{1/4} \times (988 - 212) \\ &= 25,144 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

The radiation heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.08) (0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4) [(988 + 460 \text{ R})^4 - (212 + 460 \text{ R})^4] \\ &= 575 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

Note that heat transfer by radiation is very small in this case because of the low emissivity of the surface and the relatively low surface temperature of the heating element. Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 25,144 + \frac{3}{4} \times 575 = 25,576 \text{ Btu/h} \cdot \text{ft}^2$$

Finally, the rate of heat transfer from the heating element to the water is determined by multiplying the heat flux by the heat transfer surface area,

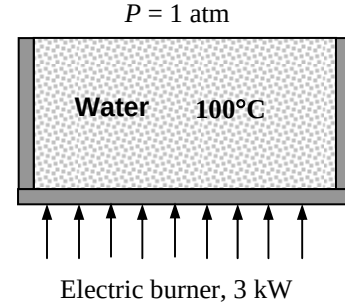
$$\begin{aligned}\dot{Q}_{\text{total}} &= A_s \dot{q}_{\text{total}} = (\pi D L) \dot{q}_{\text{total}} \\ &= (\pi \times 0.5/12 \text{ ft} \times 1 \text{ ft}) (25,576 \text{ Btu/h} \cdot \text{ft}^2) \\ &= \mathbf{3348 \text{ Btu/h}}\end{aligned}$$

10-16 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a mechanically polished AISI 304 stainless steel pan placed on top of a 3-kW electric burner. Only 60% of the heat (1.8 kW) generated is transferred to the water. The inner surface temperature of the pan and the temperature difference across the bottom of the pan are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later). 4 Heat transfer through the bottom of the pan is one-dimensional.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $k_{\text{steel}} = 14.9 \text{ W/m}\cdot^\circ\text{C}$ (Table A-3), $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\begin{aligned}\dot{Q} &= 0.60 \times 3 \text{ kW} = 1.8 \text{ kW} = 1800 \text{ W} \\ A_s &= \pi D^2 / 4 = \pi (0.30 \text{ m})^2 / 4 = 0.07069 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (1800 \text{ W}) / (0.07069 \text{ m}^2) = 25.46 \text{ W/m}^2\end{aligned}$$

Then temperature difference across the bottom of the pan is determined directly from the steady one-dimensional heat conduction relation to be

$$\dot{q} = k_{\text{steel}} \frac{\Delta T}{L} \rightarrow \Delta T = \frac{\dot{q}L}{k_{\text{steel}}} = \frac{(25,460 \text{ W/m}^2)(0.006 \text{ m})}{14.9 \text{ W/m}\cdot^\circ\text{C}} = \mathbf{10.3^\circ\text{C}}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 25,460 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = \mathbf{105.7^\circ\text{C}}$$

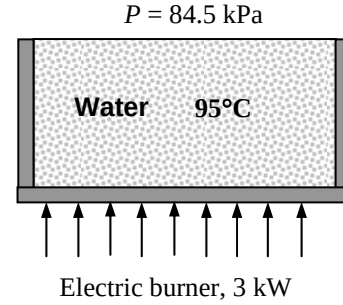
which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-17 Water is boiled at 84.5 kPa pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 95^\circ\text{C}$ in a mechanically polished AISI 304 stainless steel pan placed on top of a 3-kW electric burner. Only 60% of the heat (1.8 kW) generated is transferred to the water. The inner surface temperature of the pan and the temperature difference across the bottom of the pan are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later). 4 Heat transfer through the bottom of the pan is one-dimensional.

Properties The properties of water at the saturation temperature of 95°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 961.5 \text{ kg/m}^3 & h_{fg} &= 2270 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.50 \text{ kg/m}^3 & \mu_l &= 0.297 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0599 \text{ N/m} & C_{pl} &= 4212 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.85\end{aligned}$$



Also, $k_{\text{steel}} = 14.9 \text{ W/m}\cdot^\circ\text{C}$ (Table A-3), $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\begin{aligned}\dot{Q} &= 0.60 \times 3 \text{ kW} = 1.8 \text{ kW} = 1800 \text{ W} \\ A_s &= \pi D^2 / 4 = \pi (0.30 \text{ m})^2 / 4 = 0.07069 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (1800 \text{ W}) / (0.07069 \text{ m}^2) = 25,460 \text{ W/m}^2 = 25.46 \text{ kW/m}^2\end{aligned}$$

Then temperature difference across the bottom of the pan is determined directly from the steady one-dimensional heat conduction relation to be

$$\dot{q} = k_{\text{steel}} \frac{\Delta T}{L} \rightarrow \Delta T = \frac{\dot{q}L}{k_{\text{steel}}} = \frac{(25,460 \text{ W/m}^2)(0.006 \text{ m})}{14.9 \text{ W/m}\cdot^\circ\text{C}} = \mathbf{10.3^\circ\text{C}}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 25,460 &= (0.297 \times 10^{-3})(2270 \times 10^3) \left[\frac{9.8(961.5 - 0.50)}{0.0599} \right]^{1/2} \left(\frac{4212(T_s - 95)}{0.0130(2270 \times 10^3)1.85} \right)^3\end{aligned}$$

It gives

$$T_s = \mathbf{100.9^\circ\text{C}}$$

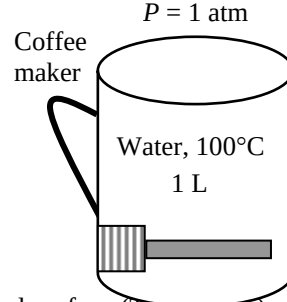
which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-18 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a stainless steel heating element. The surface temperature of the heating element and its power rating are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the coffee maker are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The density of water at room temperature is very nearly 1 kg/L , and thus the mass of 1 L water at 18°C is nearly 1 kg . The rate of energy transfer needed to evaporate half of this water in 25 min and the heat flux are

$$\begin{aligned}Q &= \dot{Q}\Delta t = mh_{fg} \rightarrow \dot{Q} = \frac{mh_{fg}}{\Delta t} = \frac{(0.5 \text{ kg})(2257 \text{ kJ/kg})}{(25 \times 60 \text{ s})} = 0.7523 \text{ kW} \\ A_s &= \pi DL = \pi(0.04 \text{ m})(0.2 \text{ m}) = 0.02513 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (0.7523 \text{ kW}) / (0.02513 \text{ m}^2) = 29.94 \text{ kW/m}^2 = 29,940 \text{ W/m}^2\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 29,940 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = 106.0^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

The specific heat of water at the average temperature of $(18+100)/2 = 59^\circ\text{C}$ is $C_p = 4.184 \text{ kJ/kg}\cdot^\circ\text{C}$. Then the time it takes for the entire water to be heated from 18°C to 100°C is determined to be

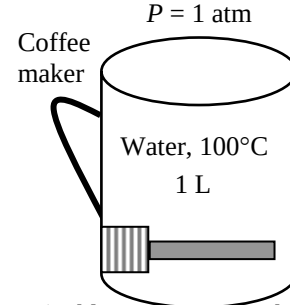
$$Q = \dot{Q}\Delta t = mC_p\Delta T \rightarrow \Delta t = \frac{mC_p\Delta T}{\dot{Q}} = \frac{(1 \text{ kg})(4.184 \text{ kJ/kg}\cdot^\circ\text{C})(100 - 18)^\circ\text{C}}{0.7523 \text{ kJ/s}} = 456 \text{ s} = 7.60 \text{ min}$$

10-19 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a copper heating element. The surface temperature of the heating element and its power rating are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the coffee maker are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a copper surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The density of water at room temperature is very nearly 1 kg/L , and thus the mass of 1 L water at 18°C is nearly 1 kg . The rate of energy transfer needed to evaporate half of this water in 25 min and the heat flux are

$$\begin{aligned}Q &= \dot{Q}\Delta t = mh_{fg} \rightarrow \dot{Q} = \frac{mh_{fg}}{\Delta t} = \frac{(0.5 \text{ kg})(2257 \text{ kJ/kg})}{(25 \times 60 \text{ s})} = 0.7523 \text{ kW} \\ A_s &= \pi DL = \pi(0.04 \text{ m})(0.2 \text{ m}) = 0.02513 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (0.7523 \text{ kW}) / (0.02513 \text{ m}^2) = 29.94 \text{ kW/m}^2 = 29,940 \text{ W/m}^2\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 29,940 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = 106.0^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

The specific heat of water at the average temperature of $(18+100)/2 = 59^\circ\text{C}$ is $C_p = 4.184 \text{ kJ/kg}\cdot^\circ\text{C}$. Then the time it takes for the entire water to be heated from 18°C to 100°C is determined to be

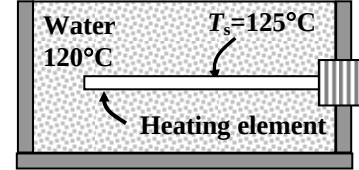
$$Q = \dot{Q}\Delta t = mC_p\Delta T \rightarrow \Delta t = \frac{mC_p\Delta T}{\dot{Q}} = \frac{(1 \text{ kg})(4.184 \text{ kJ/kg}\cdot^\circ\text{C})(100 - 18)^\circ\text{C}}{0.7523 \text{ kJ/s}} = 456 \text{ s} = 7.60 \text{ min}$$

10-20 Water is boiled at a saturation (or boiling) temperature of $T_{\text{sat}} = 120^\circ\text{C}$ by a brass heating element whose temperature is not to exceed $T_s = 125^\circ\text{C}$. The highest rate of steam production is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 125 - 120 = 5^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 943.4 \text{ kg/m}^3 & h_{fg} &= 2203 \times 10^3 \text{ J/kg} \\ \rho_v &= 1.12 \text{ kg/m}^3 & \mu_l &= 0.232 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0550 \text{ N/m} & C_{pl} &= 4244 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.44\end{aligned}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a brass surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis Assuming nucleate boiling, the heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.8(943.4 - 1.12)}{0.0550} \right]^{1/2} \left(\frac{4244(125 - 120)}{0.0060(2203 \times 10^3)1.44} \right)^3 \\ &= 290,190 \text{ W/m}^2\end{aligned}$$

The surface area of the heater is

$$A_s = \pi DL = \pi(0.02 \text{ m})(0.65 \text{ m}) = 0.04084 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.04084 \text{ m}^2)(290,190 \text{ W/m}^2) = 11,852 \text{ W}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{11,852 \text{ J/s}}{2203 \times 10^3 \text{ J/kg}} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = \mathbf{19.4 \text{ kg/h}}$$

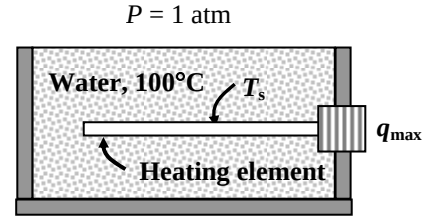
Therefore, steam can be produced at a rate of about 20 kg/h by this heater.

10-21 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a horizontal nickel plated copper heating element. The maximum (critical) heat flux and the temperature jump of the wire when the operating point jumps from nucleate boiling to film boiling regime are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel plated surface (Table 10-3). Note that we expressed the properties in units specified under Eqs. 10-2 and 10-3 in connection with their definitions in order to avoid unit manipulations. The vapor properties at the anticipated film temperature of $T_f = (T_s + T_{\text{sat}})/2$ of 100°C (will be checked) (Table A-16)

$$\begin{aligned}\rho_v &= 0.1725 \text{ kg/m}^3 \\ C_{pv} &= 2471 \text{ J/kg}\cdot^\circ\text{C} \\ k_v &= 0.1362 \text{ W/m}\cdot^\circ\text{C} \\ \mu_v &= 4.762 \times 10^{-5} \text{ kg/m}\cdot\text{s}\end{aligned}$$

Analysis (a) For a horizontal heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.0015) \left(\frac{9.8(957.9 - 0.60)}{0.0589} \right)^{1/2} = 0.60 < 1.2 \\ C_{cr} &= 0.12 L^{*-0.25} = 0.12(0.60)^{-0.25} = 0.136\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.136(2257 \times 10^3) [0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= \mathbf{1,153,000 \text{ W/m}^2}\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into the Rohsenow relation together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,153,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0060} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = 109.3^\circ\text{C}$$

(b) Heat transfer in the film boiling region can be expressed as

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 0.62 \left[\frac{gk_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4C_{pv}(T_s - T_{sat})]}{\mu_v D(T_s - T_{sat})} \right]^{1/4} (T_s - T_{sat}) + \frac{3}{4} \varepsilon \sigma (T_s^4 - T_{sat}^4)$$

Substituting,

$$1,153,000 = 0.62 \left[\frac{9.81(0.1362)^3 (0.1723)(957.9 - 0.1725)[2257 \times 10^3 + 0.4 \times 2471(T_s - 100)]}{(4.762 \times 10^{-5})(0.003)(T_s - 100)} \right]^{1/4} \times (T_s - 100) \\ + (0.5)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_s + 273)^4 - (100 + 273)^4]$$

Solving for the surface temperature gives $T_s = 1871^\circ\text{C}$. Therefore, the temperature jump of the wire when the operating point jumps from nucleate boiling to film boiling is

Temperature jump: $\Delta T = T_{s, \text{film}} - T_{s, \text{crit}} = 1871 - 109 = \mathbf{1762^\circ\text{C}}$

Note that the film temperature is $(1871+100)/2=985^\circ\text{C}$, which is close enough to the assumed value of 1000°C for the evaluation of vapor properties.

10-22 "PROBLEM 10-22"

"GIVEN"

L=0.3 "[m]"

D=0.003 "[m]"

"epsilon=0.5 parameter to be varied"

P=101.3 "[kPa], parameter to be varied"

"PROPERTIES"

Fluid\$='steam_NBS'

T_sat=temperature(Fluid\$, P=P, x=1)

rho_l=density(Fluid\$, T=T_sat, x=0)

rho_v=density(Fluid\$, T=T_sat, x=1)

sigma=SurfaceTension(Fluid\$, T=T_sat)

mu_l=Viscosity(Fluid\$, T=T_sat, x=0)

Pr_l=Prandtl(Fluid\$, T=T_sat, P=P)

C_l=CP(Fluid\$, T=T_sat, x=0)*Convert(kJ/kg-C, J/kg-C)

h_f=enthalpy(Fluid\$, T=T_sat, x=0)

h_g=enthalpy(Fluid\$, T=T_sat, x=1)

h_fg=(h_g-h_f)*Convert(kJ/kg, J/kg)

C_sf=0.0060 "from Table 8-3 of the text"

n=1 "from Table 8-3 of the text"

T_vapor=1000-273 "[C], assumed vapor temperature in the film boiling region"

rho_v_f=density(Fluid\$, T=T_vapor, P=P) "f stands for film"

C_v_f=CP(Fluid\$, T=T_vapor, P=P)*Convert(kJ/kg-C, J/kg-C)

k_v_f=Conductivity(Fluid\$, T=T_vapor, P=P)

mu_v_f=Viscosity(Fluid\$, T=T_vapor, P=P)

g=9.8 "[m/s^2], gravitational acceleration"

sigma_rad=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"

"ANALYSIS"

"(a)"

"C_cr is to be determined from Table 8-4 of the text"

C_cr=0.12*L_star^(-0.25)

L_star=D/2*((g*(rho_l-rho_v))/sigma)^0.5

q_dot_max=C_cr*h_fg*(sigma*g*rho_v^2*(rho_l-rho_v))^0.25

q_dot_nucleate=q_dot_max

q_dot_nucleate=mu_l*h_fg*(((g*(rho_l-rho_v))/sigma)^0.5)*((C_l*(T_s_crit-T_sat))/(C_sf*h_fg*Pr_l^n))^3

"(b)"

q_dot_total=q_dot_film+3/4*q_dot_rad "Heat transfer in the film boiling region"

q_dot_total=q_dot_nucleate

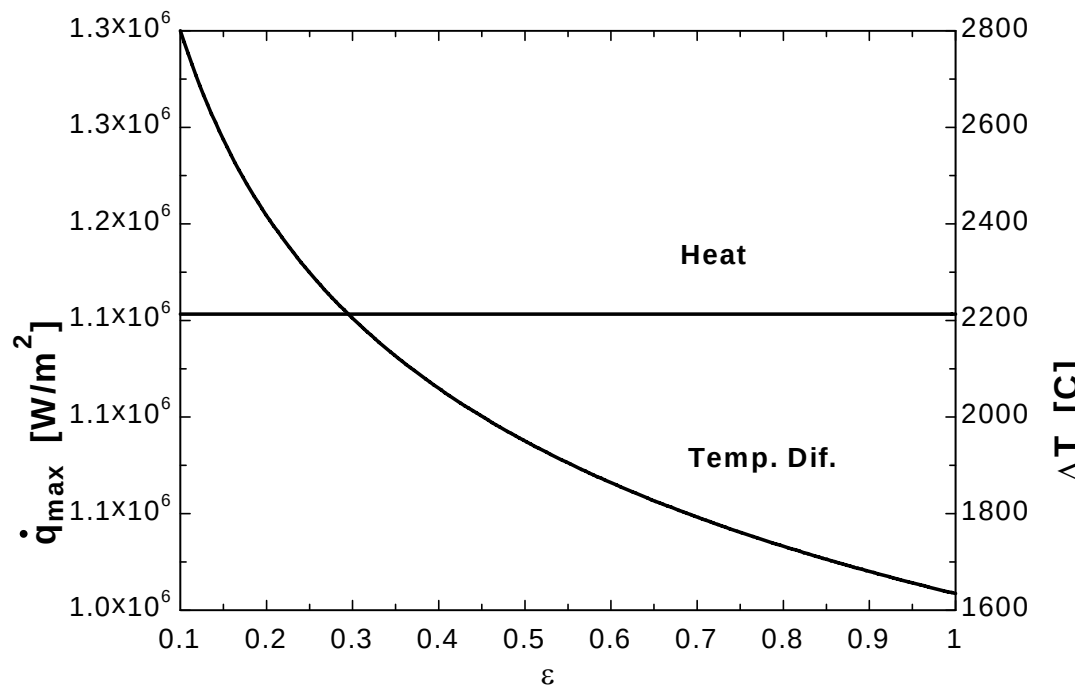
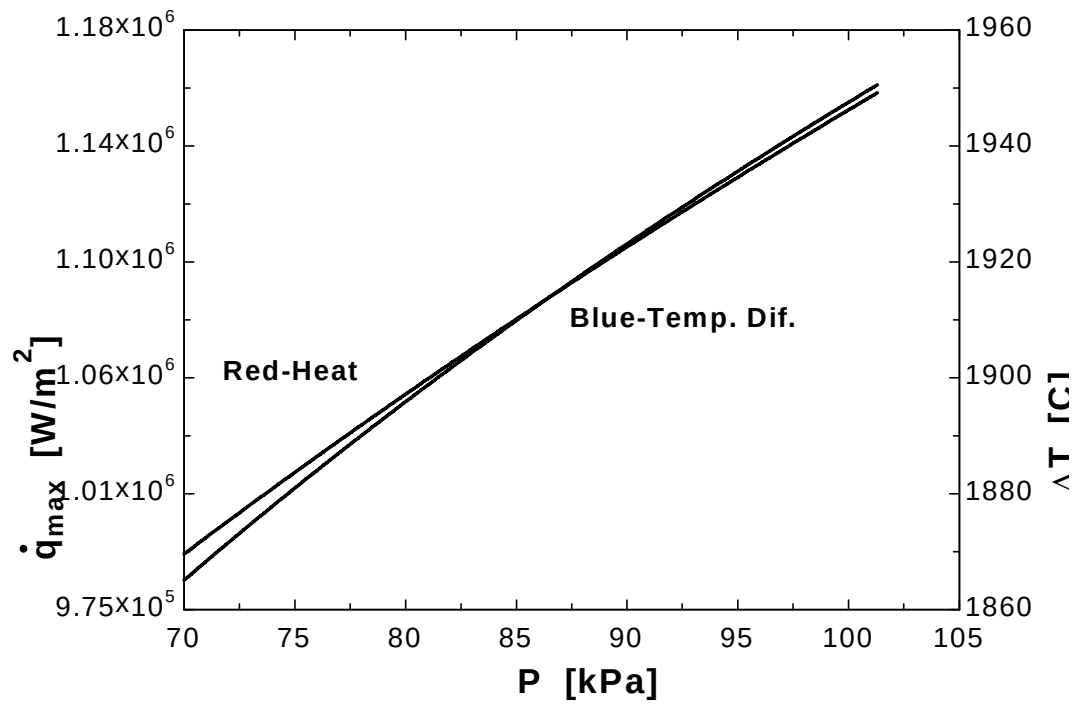
q_dot_film=0.62*((g*k_v_f^3*rho_v_f*(rho_l-rho_v_f)*(h_fg+0.4*C_v_f*(T_s_film-T_sat)))/(mu_v_f*D*(T_s_film-T_sat)))^0.25*(T_s_film-T_sat)

q_dot_rad=epsilon*sigma_rad*((T_s_film+273)^4-(T_sat+273)^4)

DELTA T=T_s_film-T_s_crit

P [kPa]	$q_{\max}$ [kW/m²]	ΔT [C]
70	994227	1865
71.65	1003642	1870
73.29	1012919	1876
74.94	1022063	1881
76.59	1031078	1886
78.24	1039970	1891
79.88	1048741	1896
81.53	1057396	1900
83.18	1065939	1905
84.83	1074373	1909
86.47	1082702	1914
88.12	1090928	1918
89.77	1099055	1923
91.42	1107085	1927
93.06	1115022	1931
94.71	1122867	1935
96.36	1130624	1939
98.01	1138294	1943
99.65	1145883	1947
101.3	1153386	1951

ε	$q_{\max}$ [kW/m²]	ΔT [C]
0.1	1153386	2800
0.15	1153386	2574
0.2	1153386	2418
0.25	1153386	2299
0.3	1153386	2205
0.35	1153386	2126
0.4	1153386	2059
0.45	1153386	2002
0.5	1153386	1951
0.55	1153386	1905
0.6	1153386	1864
0.65	1153386	1827
0.7	1153386	1793
0.75	1153386	1761
0.8	1153386	1732
0.85	1153386	1705
0.9	1153386	1680
0.95	1153386	1657
1	1153386	1634

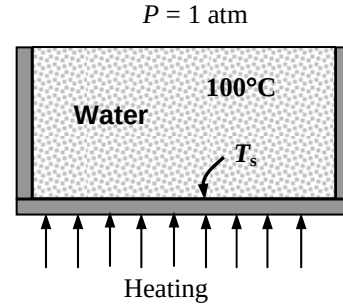


10-23 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a teflon-pitted stainless steel pan placed on an electric burner. The water level drops by 10 cm in 30 min during boiling. The inner surface temperature of the pan is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the pan are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0058$ and $n = 1.0$ for the boiling of water on a teflon-pitted stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\begin{aligned}\dot{m}_{\text{evap}} &= \frac{m_{\text{evap}}}{\Delta t} = \frac{\rho \Delta V}{\Delta t} = \frac{(957.9 \text{ kg/m}^3)(\pi \times 0.2 \text{ m} \times 0.10 \text{ m})}{30 \times 60 \text{ s}} = 0.03344 \text{ kg/s} \\ \dot{Q} &= \dot{m}_{\text{evap}} h_{fg} = (0.03344 \text{ kg/s})(2257 \text{ kJ/kg}) = 75.47 \text{ kW} \\ A_s &= \pi D^2 / 4 = \pi (0.20 \text{ m})^2 / 4 = 0.03142 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (75,470 \text{ W}) / (0.03142 \text{ m}^2) = 2,402,000 \text{ W/m}^2\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 2,402,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0058(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = 111.5^\circ\text{C}$$

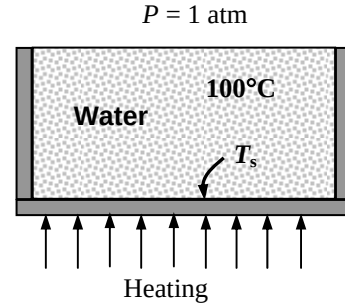
which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-24 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a polished copper pan placed on an electric burner. The water level drops by 10 cm in 30 min during boiling. The inner surface temperature of the pan is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the pan are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a copper surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\begin{aligned}\dot{m}_{\text{evap}} &= \frac{m_{\text{evap}}}{\Delta t} = \frac{\rho \Delta V}{\Delta t} = \frac{(957.9 \text{ kg/m}^3)(\pi \times 0.2 \text{ m} \times 0.10 \text{ m})}{30 \times 60 \text{ s}} = 0.03344 \text{ kg/s} \\ \dot{Q} &= \dot{m}_{\text{evap}} h_{fg} = (0.03344 \text{ kg/s})(2257 \text{ kJ/kg}) = 75.47 \text{ kW} \\ A_s &= \pi D^2 / 4 = \pi (0.20 \text{ m})^2 / 4 = 0.03142 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (75,470 \text{ W}) / (0.03142 \text{ m}^2) = 2,402,000 \text{ W/m}^2\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 2,402,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = 125.7^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-25 Water is boiled at a temperature of $T_{\text{sat}} = 150^\circ\text{C}$ by hot gases flowing through a mechanically polished stainless steel pipe submerged in water whose outer surface temperature is maintained at $T_s = 165^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation, the ratio of critical heat flux to current heat flux, and the pipe surface temperature at critical heat flux conditions are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 165 - 150 = 15^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 150°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 916.6 \text{ kg/m}^3 & h_{fg} &= 2114 \times 10^3 \text{ J/kg} \\ \rho_v &= 2.55 \text{ kg/m}^3 & \mu_l &= 0.183 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0488 \text{ N/m} & C_{pl} &= 4311 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.16\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.183 \times 10^{-3})(2114 \times 10^3) \left[\frac{9.8(916.6 - 2.55)}{0.0488} \right]^{1/2} \left(\frac{4311(165 - 150)}{0.0130(2114 \times 10^3)1.16} \right)^3 \\ &= 1,383,000 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area is

$$A_s = \pi DL = \pi(0.05 \text{ m})(50 \text{ m}) = 7.854 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (7.854 \text{ m}^2)(1,383,000 \text{ W/m}^2) = \mathbf{10,865,000 \text{ W}}$$

(b) The rate of evaporation of water is determined from

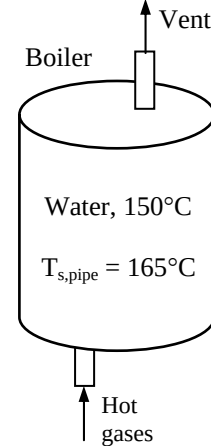
$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{10,865 \text{ kJ/s}}{2114 \text{ kJ/kg}} = \mathbf{5.139 \text{ kg/s}}$$

(c) For a horizontal cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.025) \left(\frac{9.8(916.6 - 2.55)}{0.0488} \right)^{1/2} = 10.7 > 0.12 \\ C_{cr} &= 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12(2114 \times 10^3) [0.0488 \times 9.8 \times (2.55)^2 (916.6 - 2.55)]^{1/4} \\ &= 1,852,000 \text{ W/m}^2\end{aligned}$$



Therefore,
$$\frac{\dot{q}_{\max}}{\dot{q}_{\text{current}}} = \frac{1,852,000}{1,383,000} = \mathbf{1.34}$$

(d) The surface temperature of the pipe at the burnout point is determined from Rohsenow relation at the critical heat flux value to be

$$\dot{q}_{\text{nucleate,cr}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_{s,cr} - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$1,852,000 = (0.183 \times 10^{-3})(2114 \times 10^3) \left[\frac{9.8(916.6 - 2.55)}{0.0488} \right]^{1/2} \left(\frac{4311(T_{s,cr} - 150)}{0.0130(2114 \times 10^3)1.16} \right)^3$$

$$T_{s,cr} = \mathbf{166.5^\circ\text{C}}$$

10-26 Water is boiled at a temperature of $T_{\text{sat}} = 160^\circ\text{C}$ by hot gases flowing through a mechanically polished stainless steel pipe submerged in water whose outer surface temperature is maintained at $T_s = 165^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation, the ratio of critical heat flux to current heat flux, and the pipe surface temperature at critical heat flux conditions are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 165 - 160 = 5^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 160°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 907.4 \text{ kg/m}^3 & h_{fg} &= 2083 \times 10^3 \text{ J/kg} \\ \rho_v &= 3.26 \text{ kg/m}^3 & \mu_l &= 0.170 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0466 \text{ N/m} & C_{pl} &= 4340 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.09\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.170 \times 10^{-3})(2083 \times 10^3) \left[\frac{9.8(907.4 - 3.26)}{0.0466} \right]^{1/2} \left(\frac{4340(165 - 160)}{0.0130(2083 \times 10^3)1.09} \right)^3 \\ &= 61,359 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area is

$$A_s = \pi DL = \pi(0.05 \text{ m})(50 \text{ m}) = 7.854 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (7.854 \text{ m}^2)(61,359 \text{ W/m}^2) = \mathbf{481,900 \text{ W}}$$

(b) The rate of evaporation of water is determined from

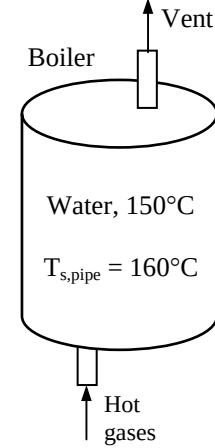
$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{481.9 \text{ kJ/s}}{2083 \text{ kJ/kg}} = \mathbf{0.231 \text{ kg/s}}$$

(c) For a horizontal cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.025) \left(\frac{9.8(907.4 - 3.26)}{0.0466} \right)^{1/2} = 10.9 > 0.12 \\ C_{cr} &= 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12(2083 \times 10^3)[0.0466 \times 9.8 \times (3.26)^2(907.4 - 3.26)]^{1/4} \\ &= 2,034,000 \text{ W/m}^2\end{aligned}$$



Therefore,
$$\frac{\dot{q}_{\max}}{\dot{q}_{\text{current}}} = \frac{2,034,000}{61,359} = \mathbf{33.2}$$

(d) The surface temperature of the pipe at the burnout point is determined from Rohsenow relation at the critical heat flux value to be

$$\dot{q}_{\text{nucleate,cr}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_{s,cr} - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$2,034,000 = (0.170 \times 10^{-3})(2083 \times 10^3) \left[\frac{9.8(907.4 - 3.26)}{0.0466} \right]^{1/2} \left(\frac{4340(T_{s,cr} - 160)}{0.0130(2083 \times 10^3)1.09} \right)^3$$

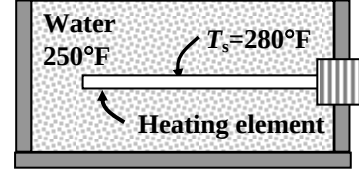
$$T_{s,cr} = \mathbf{176.1^\circ\text{C}}$$

10-27E Water is boiled at a temperature of $T_{\text{sat}} = 250^\circ\text{F}$ by a nickel-plated heating element whose surface temperature is maintained at $T_s = 280^\circ\text{F}$. The boiling heat transfer coefficient, the electric power consumed, and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 280 - 250 = 30^\circ\text{F}$ which is in the nucleate boiling range of 9 to 55°F for water.

Properties The properties of water at the saturation temperature of 250°F are (Tables 10-1 and A-9E)

$$\begin{aligned}\rho_l &= 58.82 \text{ lbm} / \text{ft}^3 & h_{fg} &= 946 \text{ Btu} / \text{lbm} \\ \rho_v &= 0.0723 \text{ lbm} / \text{ft}^3 & \mu_l &= 0.556 \text{ lbm} / \text{h} \cdot \text{ft} \\ \sigma &= 0.003755 \text{ lbf} / \text{ft} = 0.1208 \text{ lbm} / \text{s}^2 & C_{pl} &= 1.015 \text{ Btu} / \text{lbm} \cdot ^\circ\text{F} \\ \text{Pr}_l &= 1.43\end{aligned}$$



Also, $g = 32.2 \text{ ft/s}^2$ and $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel plated surface (Table 10-3). Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.556)(946) \left[\frac{32.2(58.82 - 0.0723)}{0.1208} \right]^{1/2} \left(\frac{1.015(280 - 250)}{0.0060(946)1.43} \right)^3 \\ &= 3,475,221 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

Then the convection heat transfer coefficient becomes

$$\dot{q} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{\dot{q}}{T_s - T_{\text{sat}}} = \frac{3,471,670 \text{ Btu} / \text{h} \cdot \text{ft}^2}{(280 - 250)^\circ\text{F}} = 115,840 \text{ Btu} / \text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$$

(b) The electric power consumed is equal to the rate of heat transfer to the water, and is determined from

$$\begin{aligned}\dot{W}_e = \dot{Q} &= \dot{Q}_s = (\pi DL)\dot{q} = (\pi \times 0.5 / 12 \text{ ft} \times 2 \text{ ft})(3,475,221 \text{ Btu/h} \cdot \text{ft}^2) = 909,811 \text{ Btu/h} \\ &= \mathbf{266.6 \text{ kW}} \quad (\text{since } 1 \text{ kW} = 3412 \text{ Btu/h})\end{aligned}$$

(c) Finally, the rate of evaporation of water is determined from

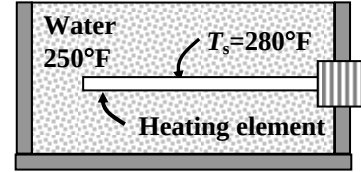
$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{909,811 \text{ Btu/h}}{946 \text{ Btu/lbm}} = \mathbf{961.7 \text{ lbm/h}}$$

10-28E Water is boiled at a temperature of $T_{\text{sat}} = 250^\circ\text{F}$ by a platinum-plated heating element whose surface temperature is maintained at $T_s = 280^\circ\text{F}$. The boiling heat transfer coefficient, the electric power consumed, and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 280 - 250 = 30^\circ\text{F}$ which is in the nucleate boiling range of 9 to 55°F for water.

Properties The properties of water at the saturation temperature of 250°F are (Tables 10-1 and A-9E)

$$\begin{aligned}\rho_l &= 58.82 \text{ lbm} / \text{ft}^3 & h_{fg} &= 946 \text{ Btu} / \text{lbm} \\ \rho_v &= 0.0723 \text{ lbm} / \text{ft}^3 & \mu_l &= 0.556 \text{ lbm} / \text{h} \cdot \text{ft} \\ \sigma &= 0.003755 \text{ lbf} / \text{ft} = 0.1208 \text{ lbm} / \text{s}^2 & C_{pl} &= 1.015 \text{ Btu} / \text{lbm} \cdot ^\circ\text{F} \\ \text{Pr}_l &= 1.43\end{aligned}$$



Also, $g = 32.2 \text{ ft/s}^2$ and $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a platinum plated surface (Table 10-3). Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.556)(946) \left[\frac{32.2(58.82 - 0.0723)}{0.1208} \right]^{1/2} \left(\frac{1.015(280 - 250)}{0.0130(0.1208 \times 10^3)^{1.43}} \right)^3 \\ &= 341,670 \text{ Btu/h} \cdot \text{ft}^2\end{aligned}$$

Then the convection heat transfer coefficient becomes

$$\dot{q} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{\dot{q}}{T_s - T_{\text{sat}}} = \frac{341,670 \text{ Btu} / \text{h} \cdot \text{ft}^2}{(280 - 250)^\circ\text{F}} = \mathbf{11,390 \text{ Btu} / \text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}}$$

(b) The electric power consumed is equal to the rate of heat transfer to the water, and is determined from

$$\begin{aligned}\dot{W}_e = \dot{Q} = \dot{Q}_s &= (\pi DL)\dot{q} = (\pi \times 0.5 / 12 \text{ ft} \times 2 \text{ ft})(341,670 \text{ Btu/h} \cdot \text{ft}^2) = 89,450 \text{ Btu/h} \\ &= \mathbf{26.2 \text{ kW}} \quad (\text{since } 1 \text{ kW} = 3412 \text{ Btu/h})\end{aligned}$$

(c) Finally, the rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{89,450 \text{ Btu/h}}{946 \text{ Btu/lbm}} = \mathbf{94.6 \text{ lbm/h}}$$

10-29E "PROBLEM 10-29E"

"GIVEN"

T_sat=250 "[F]"

L=2 "[ft]"

D=0.5/12 "[ft]"

"T_s=280 [F], parameter to be varied"

"PROPERTIES"

Fluid\$='steam_NBS'

P_sat=pressure(Fluid\$, T=T_sat, x=1)

rho_l=density(Fluid\$, T=T_sat, x=0)

rho_v=density(Fluid\$, T=T_sat, x=1)

sigma=SurfaceTension(Fluid\$, T=T_sat)*Convert(lbf/ft, lbm/s^2)

mu_l=Viscosity(Fluid\$, T=T_sat, x=0)

Pr_l=Prandtl(Fluid\$, T=T_sat, P=P_sat+1) "P=P_sat+1 is used to get liquid state"

C_l=CP(Fluid\$, T=T_sat, x=0)

h_f=enthalpy(Fluid\$, T=T_sat, x=0)

h_g=enthalpy(Fluid\$, T=T_sat, x=1)

h_fg=h_g-h_f

C_sf=0.0060 "from Table 8-3 of the text"

n=1 "from Table 8-3 of the text"

g=32.2 "[ft/s^2], gravitational acceleration"

"ANALYSIS"**"(a)"**

$$q_{\text{dot_nucleate}} = \mu_l h_{fg} \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{0.5} \left(C_l (T_s - T_{\text{sat}}) \right) / (C_{sf} h_{fg} \text{Pr}_l^n)^{1/3}$$

$$q_{\text{dot_nucleate}} = h (T_s - T_{\text{sat}})$$
"(b)"

W_dot_e=q_dot_nucleate*A*Convert(Btu/h, kW)

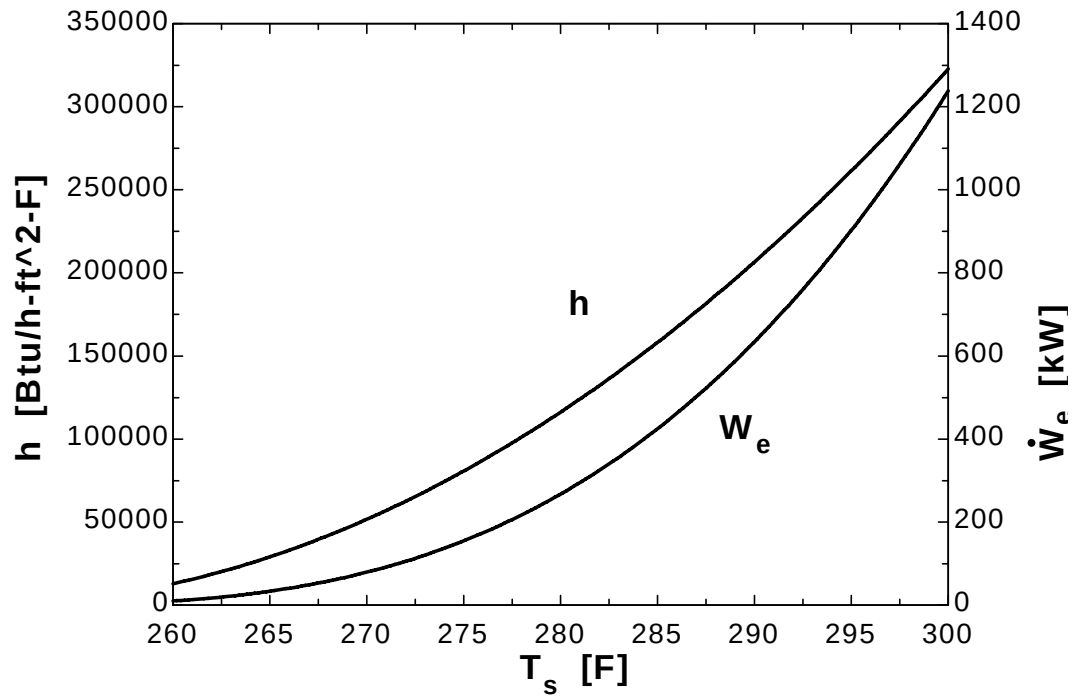
A=pi*D*L

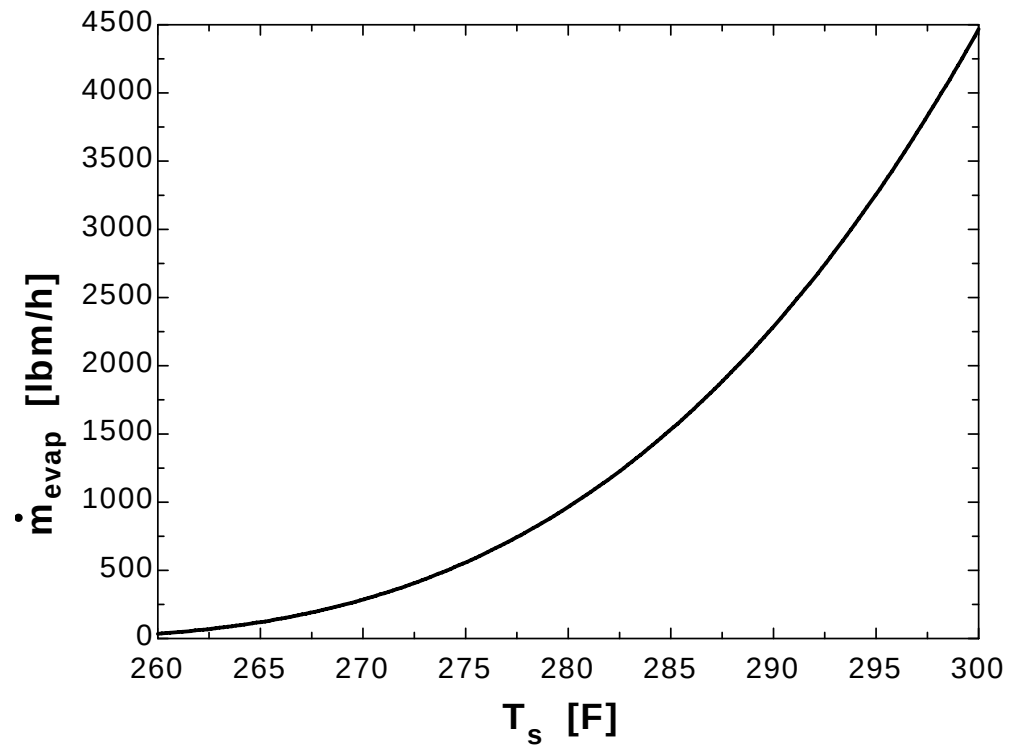
"(c)"

m_dot_evap=Q_dot_boiling/h_fg

Q_dot_boiling=W_dot_e*Convert(kW, Btu/h)

T_s [F]	h [Btu/h.ft ² .F]	$\dot{W}_e$ [kW]	$\dot{m}_{\text{evap}}$ [lbm/h]
260	12908	9.903	35.74
262	18587	17.11	61.76
264	25299	27.18	98.07
266	33043	40.56	146.4
268	41821	57.76	208.4
270	51630	79.23	285.9
272	62473	105.5	380.5
274	74348	136.9	494.1
276	87255	174.1	628.1
278	101195	217.4	784.5
280	116168	267.4	964.9
282	132174	324.5	1171
284	149212	389.2	1405
286	167282	462.1	1667
288	186386	543.4	1961
290	206521	633.8	2287
292	227690	733.7	2648
294	249891	843.6	3044
296	273125	964	3479
298	297391	1095	3952
300	322690	1238	4467





10-30 Cold water enters a steam generator at 15°C and is boiled, and leaves as saturated vapor at $T_{\text{sat}} = 100^\circ\text{C}$. The fraction of heat used to preheat the liquid water from 15°C to saturation temperature of 100°C is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the steam generator are negligible.

Properties The heat of vaporization of water at 100°C is $h_{\text{fg}} = 2257 \text{ kJ/kg}$ and the specific heat of liquid water at the average temperature of $(15+100)/2 = 57.5^\circ\text{C}$ is $C_{pl} = 4.184 \text{ kJ/kg} \cdot ^\circ\text{C}$ (Table A-9).

Analysis The heat of vaporization of water represents the amount of heat needed to vaporize a unit mass of liquid at a specified temperature. Using the average specific heat, the amount of heat needed to preheat a unit mass of water from 15°C to 100°C is determined to be

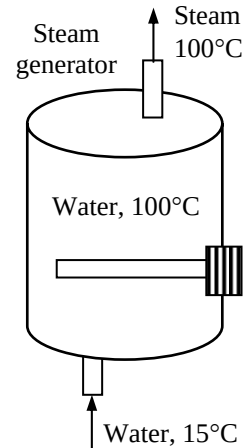
$$q_{\text{preheating}} = C_{pl} \Delta T = (4.184 \text{ kJ/kg} \cdot ^\circ\text{C})(100 - 15)^\circ\text{C} = 355.6 \text{ kJ/kg}$$

and

$$q_{\text{total}} = q_{\text{boiling}} + q_{\text{preheating}} = 2257 + 355.6 = 2612.6 \text{ kJ/kg}$$

Therefore, the fraction of heat used to preheat the water is

$$\text{Fraction to preheat} = \frac{q_{\text{preheating}}}{q_{\text{total}}} = \frac{355.6}{2612.6} = \mathbf{0.136} \text{ (or } \mathbf{13.6\%})$$



10-31 Cold water enters a steam generator at 20°C and is boiled, and leaves as saturated vapor at boiler pressure. The boiler pressure at which the amount of heat needed to preheat the water to saturation temperature is equal to the heat of vaporization is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the steam generator are negligible.

Properties The properties needed to solve this problem are the heat of vaporization h_{fg} and the specific heat of water C_p at specified temperatures, and they can be obtained from Table A-9.

Analysis The heat of vaporization of water represents the amount of heat needed to vaporize a unit mass of liquid at a specified temperature, and $C_p \Delta T$ represents the amount of heat needed to preheat a unit mass of water from 20°C to the saturation temperature. Therefore,

$$q_{\text{preheating}} = q_{\text{boiling}}$$

$$C_{p,ave}(T_{\text{sat}} - 20) = h_{fg@T_{\text{sat}}}$$

The solution of this problem requires choosing a boiling temperature, reading the heat of vaporization at that temperature, evaluating the specific heat at the average temperature, and substituting the values into the relation above to see if it is satisfied. By trial and error, the temperature that satisfies this condition is determined to be 315°C at which (Table A-9)

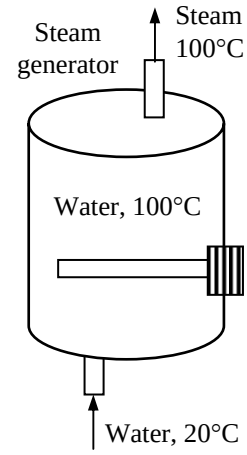
$$h_{fg@315^\circ\text{C}} = 1281 \text{ kJ/kg} \quad \text{and} \quad T_{\text{ave}} = (20 + 315)/2 = 167.5^\circ\text{C} \rightarrow C_{p,ave} = 4.37 \text{ kJ/kg}\cdot^\circ\text{C}$$

Substituting,

$$C_{p,ave}(T_{\text{sat}} - 20) = (4.37 \text{ kJ/kg}\cdot^\circ\text{C})(315 - 20)^\circ\text{C} = 1289 \text{ kJ/kg}$$

which is practically identical to the heat of vaporization. Therefore,

$$P_{\text{boiler}} = P_{\text{sat}@T_{\text{sat}}} = \mathbf{10.6 \text{ MPa}}$$

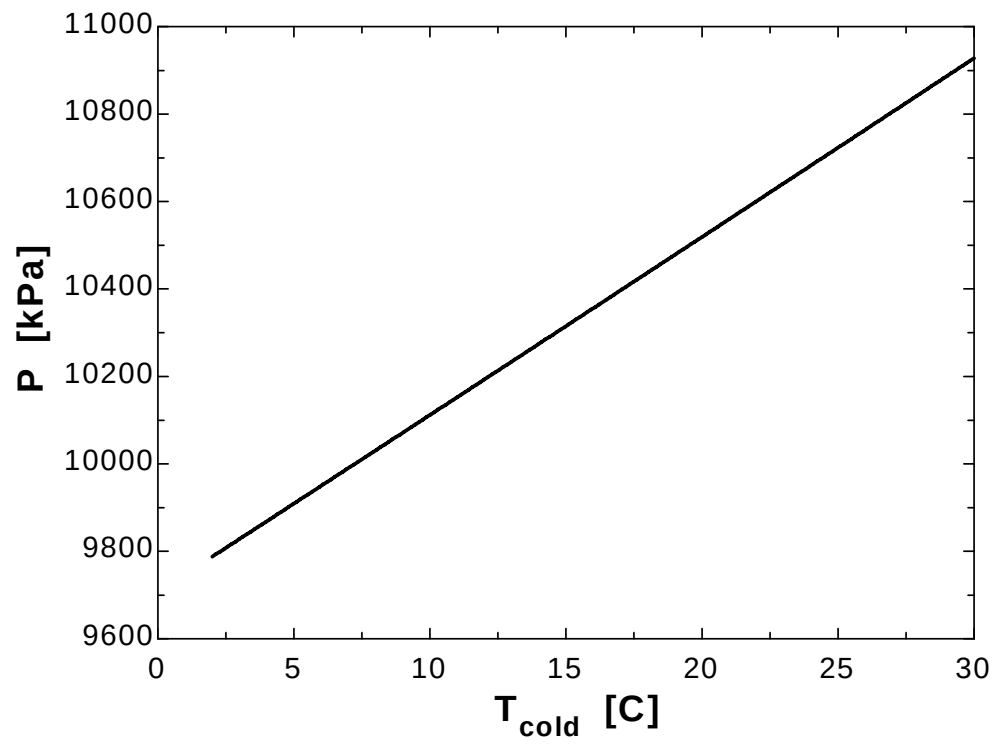


10-32 "PROBLEM 10-32"**"GIVEN"****"T_{cold}=20 [C], parameter to be varied"****"ANALYSIS"**

Fluid\$='steam_NBS'

q_{preheating}=q_{boiling}q_{preheating}=C_p*(T_{sat}-T_{cold})T_{sat}=temperature(Fluid\$, P=P, x=1)C_p=CP(Fluid\$, T=T_{ave}, x=0)T_{ave}=1/2*(T_{cold}+T_{sat})q_{boiling}=h_{fg}h_f=enthalpy(Fluid\$, T=T_{sat}, x=0)h_g=enthalpy(Fluid\$, T=T_{sat}, x=1)h_{fg}=h_g-h_f

T _{cold} [C]	P [kPa]
8	10031
9	10071
10	10112
11	10152
12	10193
13	10234
14	10274
15	10315
16	10356
17	10396
18	10437
19	10478
20	10519
21	10560
22	10601
23	10641
24	10682
25	10723
26	10764
27	10805
28	10846
29	10887
30	10928



10-33 Boiling experiments are conducted by heating water at 1 atm pressure with an electric resistance wire, and measuring the power consumed by the wire as well as temperatures. The boiling heat transfer coefficient is to be determined.

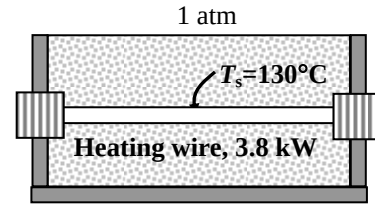
Assumptions 1 Steady operating conditions exist. 2 Heat losses from the water are negligible.

Analysis The heat transfer area of the heater wire is

$$A_s = \pi DL = \pi(0.002 \text{ m})(0.50 \text{ m}) = 0.003142 \text{ m}^2$$

Noting that 3800 W of electric power is consumed when the heater surface temperature is 130°C, the boiling heat transfer coefficient is determined from Newton's law of cooling to be

$$\dot{Q} = hA_s(T_s - T_{\text{sat}}) \rightarrow h = \frac{\dot{Q}}{A_s(T_s - T_{\text{sat}})} = \frac{3800 \text{ W}}{(0.003142 \text{ m}^2)(130 - 100)^\circ\text{C}} = 40,320 \text{ W/m}^2 \cdot ^\circ\text{C}$$



Condensation Heat Transfer

10-34C Condensation is a vapor-to-liquid phase change process. It occurs when the temperature of a vapor is reduced *below* its saturation temperature T_{sat} . This is usually done by bringing the vapor into contact with a solid surface whose temperature T_s is *below* the saturation temperature T_{sat} of the vapor.

10-35C In *film condensation*, the condensate wets the surface and forms a liquid film on the surface which slides down under the influence of gravity. The thickness of the liquid film increases in the flow direction as more vapor condenses on the film. This is how condensation normally occurs in practice. In *dropwise condensation*, the condensed vapor forms droplets on the surface instead of a continuous film, and the surface is covered by countless droplets of varying diameters. Dropwise condensation is a much more effective mechanism of heat transfer.

10-36C In condensate flow, the wetted perimeter is defined as the length of the surface-condensate interface at a cross-section of condensate flow. It differs from the ordinary perimeter in that the latter refers to the entire circumference of the condensate at some cross-section.

10-37C The modified latent heat of vaporization h_{fg}^* is the amount of heat released as a unit mass of vapor condenses at a specified temperature, plus the amount of heat released as the condensate is cooled further to some average temperature between T_{sat} and T_s . It is defined as $h_{fg}^* = h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s)$ where C_{pl} is the specific heat of the liquid at the average film temperature.

10-38C During film condensation on a vertical plate, heat flux at the top will be higher since the thickness of the film at the top, and thus its thermal resistance, is lower.

10-39C Setting the heat transfer coefficient relations for a vertical tube of height L and a horizontal tube of diameter D equal to each other yields $L = 2.77D$, which implies that for a tube whose length is 2.77 times its diameter, the average heat transfer coefficient for laminar film condensation will be the *same* whether the tube is positioned horizontally or vertically. For $L = 10D$, the heat transfer coefficient and thus the heat transfer rate will be higher in the horizontal position since $L > 2.77D$ in that case.

10-40C The condensation heat transfer coefficient for the tubes will be the highest for the case of horizontal side by side (case b) since (1) for long tubes, the horizontal position gives the highest heat transfer coefficients, and (2) for tubes in a vertical tier, the average thickness of the liquid film at the lower tubes is much larger as a result of condensate falling on top of them from the tubes directly above, and thus the average heat transfer coefficient at the lower tubes in such arrangements is smaller.

10-41C The presence of noncondensable gases in the vapor has a detrimental effect on condensation heat transfer. Even small amounts of a noncondensable gas in the vapor cause significant drops in heat transfer coefficient during condensation.

10-42 The hydraulic diameter D_h for all 4 cases are expressed in terms of the boundary layer thickness δ as follows:

$$(a) \text{ Vertical plate: } D_h = \frac{4A_c}{p} = \frac{4w\delta}{w} = 4\delta$$

$$(b) \text{ Tilted plate: } D_h = \frac{4A_c}{p} = \frac{4w\delta}{w} = 4\delta$$

$$(c) \text{ Vertical cylinder: } D_h = \frac{4A_c}{p} = \frac{4\pi D\delta}{\pi D} = 4\delta$$

$$(d) \text{ Horizontal cylinder: } D_h = \frac{4A_c}{p} = \frac{4(2L\delta)}{2L} = 4\delta$$

$$(e) \text{ Sphere: } D_h = \frac{4A_c}{p} = \frac{4\pi D\delta}{\pi D} = 4\delta$$

Therefore, the Reynolds number for all 5 cases can be expressed as

$$\text{Re} = \frac{4\dot{m}}{p\mu_l} = \frac{4A_c\rho_l V_l}{p\mu_l} = \frac{D_h\rho_l V_l}{\mu_l} = \frac{4\delta\rho_l V_l}{\mu_l}$$

10-43 There is film condensation on the outer surfaces of N horizontal tubes arranged in a vertical tier. The value of N for which the average heat transfer coefficient for the entire tier be equal to half of the value for a single horizontal tube is to be determined.

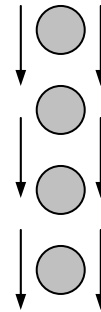
Assumptions Steady operating conditions exist.

Analysis The relation between the heat transfer coefficients for the two cases is given to be

$$h_{\text{horizontal, N tubes}} = \frac{h_{\text{horizontal, 1 tube}}}{N^{1/4}}$$

Therefore,

$$\frac{h_{\text{horizontal, N tubes}}}{h_{\text{horizontal, 1 tube}}} = \frac{1}{2} = \frac{1}{N^{1/4}} \longrightarrow N = \mathbf{16}$$

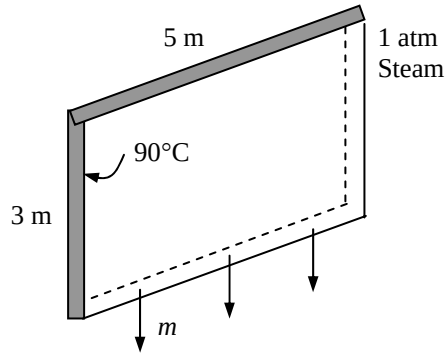


10-44 Saturated steam at atmospheric pressure thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a vertical plate which is maintained at 90°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (100 + 90) / 2 = 95^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 961.5 \text{ kg/m}^3 \\ \mu_l &= 0.297 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.309 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4212 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.677 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4212 \text{ J/kg}\cdot^\circ\text{C}(100 - 90)^\circ\text{C} = 2,286 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} = \text{Re}_{\text{vertical, wavy}} &= \left[4.81 + \frac{3.70Lk_l(T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (3 \text{ m}) \times (0.677 \text{ W/m}\cdot^\circ\text{C}) \times (100 - 90)^\circ\text{C}}{(0.297 \times 10^{-3} \text{ kg/m}\cdot\text{s}) (2286 \times 10^3 \text{ J/kg})} \left(\frac{9.8 \text{ m/s}^2}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 1112\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{1112 \times (0.677 \text{ W/m}\cdot^\circ\text{C})}{1.08(1112)^{1.22} - 5.2} \left(\frac{9.8 \text{ m/s}^2}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 6279 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is

$$A_s = W \times L = (3 \text{ m})(5 \text{ m}) = 15 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (6279 \text{ W/m}^2\cdot^\circ\text{C})(15 \text{ m}^2)(100 - 90)^\circ\text{C} = \mathbf{941,850 \text{ W}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{941,850 \text{ J/s}}{2286 \times 10^3 \text{ J/kg}} = \mathbf{0.412 \text{ kg/s}}$$

10-45 Saturated steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a plate which is tilted 60° from the vertical and maintained at 90°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of the steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 90)/2 = 95^\circ\text{C}$ are (Table A-9),

$$\rho_l = 961.5 \text{ kg/m}^3$$

$$\mu_l = 0.297 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.309 \times 10^{-6} \text{ m}^2/\text{s}$$

$$C_{pl} = 4212 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.677 \text{ W/m}\cdot^\circ\text{C}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4212 \text{ J/kg}\cdot^\circ\text{C} (100 - 90)^\circ\text{C} \\ &= 2,286 \times 10^3 \text{ J/kg} \end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from the vertical plate relation by replacing g by $g \cos \theta$ where $\theta = 60^\circ$ to be

$$\begin{aligned} \text{Re} &= \text{Re}_{\text{tilted, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g \cos 60^\circ}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (3 \text{ m}) \times (0.677 \text{ W/m}\cdot^\circ\text{C}) \times (100 - 90)^\circ\text{C}}{(0.297 \times 10^{-3} \text{ kg/m}\cdot\text{s}) (2286 \times 10^3 \text{ J/kg})} \left(\frac{(9.8 \text{ m/s}^2) \cos 60^\circ}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 950.5 \end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{tilted, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \\ &= \frac{950.5 \times (0.677 \text{ W/m}\cdot^\circ\text{C})}{1.08(950.5)^{1.22} - 5.2} \left(\frac{(9.8 \text{ m/s}^2) \cos 60^\circ}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5159 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The heat transfer surface area of the plate is $A_s = W \times L = (3 \text{ m})(5 \text{ m}) = 15 \text{ m}^2$.

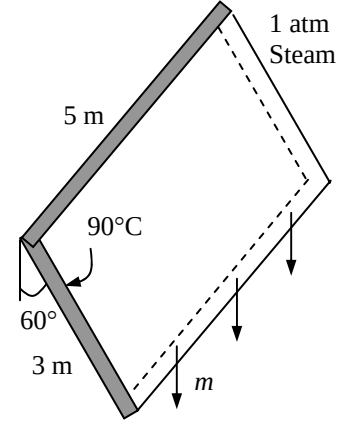
Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (5159 \text{ W/m}^2\cdot^\circ\text{C})(15 \text{ m}^2)(100 - 90)^\circ\text{C} = \mathbf{773,850 \text{ W}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{773,850 \text{ J/s}}{2286 \times 10^3 \text{ J/kg}} = \mathbf{0.339 \text{ kg/s}}$$

Discussion Using the heat transfer coefficient determined in the previous problem for the vertical plate, we could also determine the heat transfer coefficient from $h_{\text{inclined}} = h_{\text{vert}} (\cos \theta)^{1/4}$. It would give $5280 \text{ W/m}^2\cdot^\circ\text{C}$, which is 2.3% different than the value determined above.



10-46 Saturated steam condenses outside of vertical tube. The rate of heat transfer to the coolant, the rate of condensation and the thickness of the condensate layer at the bottom are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The tube can be treated as a vertical plate. 4 The condensate flow is wavy-laminar over the entire tube (this assumption will be verified). 5 Nusselt's analysis can be used to determine the thickness of the condensate film layer. 6 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 30°C are $h_{fg} = 2431 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.03 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (30 + 20) / 2 = 25^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 997.0 \text{ kg/m}^3 \\ \mu_l &= 1.002 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 1.005 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4180 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.607 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2431 \times 10^3 \text{ J/kg} + 0.68 \times 4180 \text{ J/kg}\cdot^\circ\text{C} (30 - 20)^\circ\text{C} = 2459 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.607 \text{ W/m}\cdot^\circ\text{C}) \times (30 - 20)^\circ\text{C}}{(1.002 \times 10^{-3} \text{ kg/m}\cdot\text{s}) (2459 \times 10^3 \text{ J/kg})} \left(\frac{9.8 \text{ m/s}^2}{(1.005 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 133.9\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{ Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{133.9 \times (0.607 \text{ W/m}\cdot^\circ\text{C})}{1.08 (133.9)^{1.22} - 5.2} \left(\frac{9.8 \text{ m/s}^2}{(1.005 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 4132 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the tube is $A_s = \pi D L = \pi (0.04 \text{ m}) (2 \text{ m}) = 0.2513 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

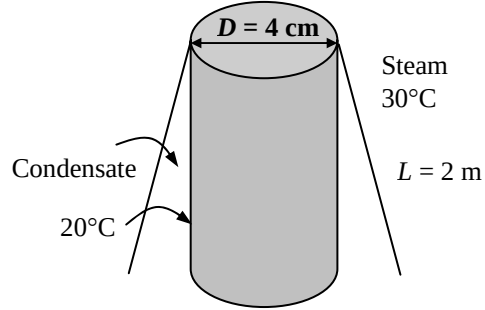
$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4132 \text{ W/m}^2 \cdot ^\circ\text{C}) (0.2513 \text{ m}^2) (30 - 20)^\circ\text{C} = \mathbf{10,385 \text{ W}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{10,385 \text{ J/s}}{2459 \times 10^3 \text{ J/kg}} = \mathbf{4.22 \times 10^{-3} \text{ kg/s}}$$

(c) Combining equations $\delta_L = k_l / h_l$ and $h = (4/3) h_L$, the thickness of the liquid film at the bottom of the tube is determined to be

$$\delta_L = \frac{4 k_l}{3 h} = \frac{4 (0.607 \text{ W/m}\cdot^\circ\text{C})}{3 (4132 \text{ W/m}^2 \cdot ^\circ\text{C})} = 0.196 \times 10^{-3} = \mathbf{0.2 \text{ mm}}$$

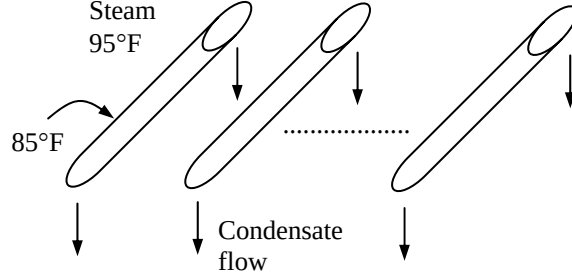


10-47E Saturated steam at a saturation temperature of $T_{\text{sat}} = 95^\circ\text{F}$ condenses on the outer surfaces of horizontal pipes which are maintained at 85°F by circulating cooling water. The rate of heat transfer to the cooling water and the rate of condensation per unit length of a single horizontal pipe are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The pipe is isothermal. 3 There is no interference between the pipes (no drip of the condensate from one tube to another).

Properties The properties of water at the saturation temperature of 95°F are $h_{\text{fg}} = 1040 \text{ Btu/lbm}$ and $\rho_v = 0.0025 \text{ lbm/ft}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (95 + 85)/2 = 90^\circ\text{F}$ are (Table A-9E),

$$\begin{aligned}\rho_l &= 62.12 \text{ lbm/ft}^3 \\ \mu_l &= 1.842 \text{ lbm/ft} \cdot \text{h} \\ \nu_l &= \mu_l / \rho_l = 0.02965 \text{ ft}^2 / \text{h} \\ C_{pl} &= 0.999 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_l &= 0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1040 \text{ Btu/lbm} + 0.68 \times (0.999 \text{ Btu/lbm} \cdot ^\circ\text{F})(95 - 85)^\circ\text{F} \\ &= 1047 \text{ Btu/lbm}\end{aligned}$$

Noting that we have condensation on a horizontal tube, the heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(32.2 \text{ ft/s}^2)(62.12 \text{ lbm/ft}^3)(62.12 - 0.0025 \text{ lbm/ft}^3)(1047 \text{ Btu/lbm})(0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F})^3}{[(1 \text{ h}/3600 \text{ s})^2](1.842 \text{ lbm/ft} \cdot \text{h})(95 - 85)^\circ\text{F}(1/12 \text{ ft})} \right]^{1/4} \\ &= 1942 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}\end{aligned}$$

The heat transfer surface area of the tube per unit length is

$$A_s = \pi DL = \pi(1/12 \text{ ft})(1 \text{ ft}) = 0.2618 \text{ ft}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (1942 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(0.2618 \text{ ft}^2)(95 - 85)^\circ\text{F} = \mathbf{5084 \text{ Btu/h}}$$

(b) The rate of condensation of steam is determined from

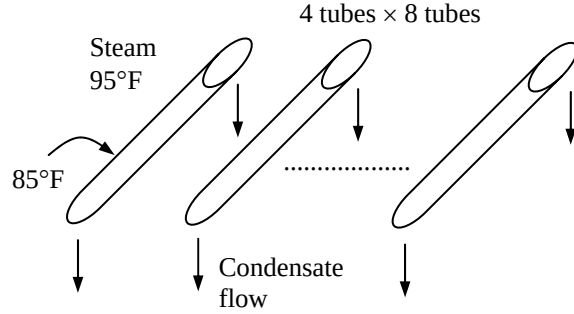
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{5084 \text{ Btu/h}}{1047 \text{ Btu/lbm}} = \mathbf{4.856 \text{ lbm/h}}$$

10-48E Saturated steam at a saturation temperature of $T_{\text{sat}} = 95^\circ\text{F}$ condenses on the outer surfaces of 20 horizontal pipes which are maintained at 85°F by circulating cooling water and arranged in a rectangular array of 4 pipes high and 5 pipes wide. The rate of heat transfer to the cooling water and the rate of condensation per unit length of the pipes are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The pipes are isothermal.

Properties The properties of water at the saturation temperature of 95°F are $h_{fg} = 1040 \text{ Btu/lbm}$ and $\rho_v = 0.0025 \text{ lbm/ft}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (95 + 85)/2 = 90^\circ\text{F}$ are (Table A-9E),

$$\begin{aligned}\rho_l &= 62.12 \text{ lbm/ft}^3 \\ \mu_l &= 1.842 \text{ lbm/ft} \cdot \text{h} \\ \nu_l &= \mu_l / \rho_l = 0.02965 \text{ ft}^2 / \text{h} \\ C_{pl} &= 0.999 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_l &= 0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1040 \text{ Btu/lbm} + 0.68 \times (0.999 \text{ Btu/lbm} \cdot ^\circ\text{F})(95 - 85)^\circ\text{F} = 1047 \text{ Btu/lbm}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal pipe is

$$\begin{aligned}h &= h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(32.2 \text{ ft/s}^2)(62.12 \text{ lbm/ft}^3)(62.12 - 0.0025 \text{ lbm/ft}^3)(1047 \text{ Btu/lbm})(0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F})^3}{[(1 \text{ h}/3600 \text{ s})^2](1.842 \text{ lbm/ft} \cdot \text{h})(95 - 85)^\circ\text{F}(1/12 \text{ ft})} \right]^{1/4} \\ &= 1942 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}\end{aligned}$$

Then the average heat transfer coefficient for a 4-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{4^{1/4}} (1942 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) = 1373 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$$

The surface area for all 32 pipes per unit length of the pipes is

$$A_s = N_{\text{total}} \pi D L = 32 \pi (1/12 \text{ ft})(1 \text{ ft}) = 8.378 \text{ ft}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (1373 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(8.378 \text{ ft}^2)(95 - 85)^\circ\text{F} = \mathbf{115,000 \text{ Btu/h}}$$

(b) The rate of condensation of steam is determined from

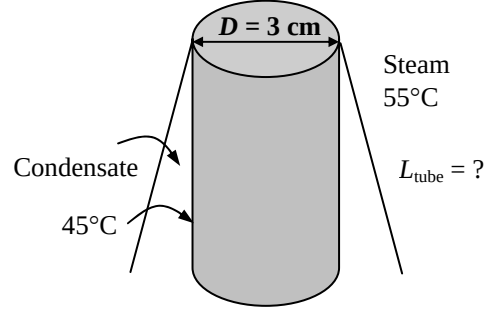
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{115,000 \text{ Btu/h}}{1047 \text{ Btu/lbm}} = \mathbf{109.9 \text{ lbm/h}}$$

10-49 Saturated steam at a saturation temperature of $T_{\text{sat}} = 55^\circ\text{C}$ condenses on the outer surface of a vertical tube which is maintained at 45°C . The required tube length to condense steam at a rate of 10 kg/h is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vertical tube can be treated as a vertical plate. 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 55°C are $h_{fg} = 2371 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.1045 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (55 + 45)/2 = 50^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 988.1 \text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4181 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.644 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 2371 \times 10^3 \text{ J/kg} + 0.68 \times 4181 \text{ J/kg}\cdot^\circ\text{C}(55 - 45)^\circ\text{C} = 2399 \times 10^3 \text{ J/kg}\end{aligned}$$

The Reynolds number is determined from its definition to be

$$\text{Re} = \frac{4\dot{m}}{p\mu_l} = \frac{4(10/3600 \text{ kg/s})}{\pi(0.03 \text{ m})(0.547 \times 10^{-3} \text{ kg/m}\cdot\text{s})} = 215.5$$

which is between 30 and 1800. Therefore the condensate flow is wavy laminar, and the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{ Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{215.5 \times (0.644 \text{ W/m}\cdot^\circ\text{C})}{1.08(215.5)^{1.22} - 5.2} \left(\frac{9.8 \text{ m/s}^2}{(0.554 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5644 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The rate of heat transfer during this condensation process is

$$\dot{Q} = \dot{m}h_{fg}^* = (10/3600 \text{ kg/s})(2399 \times 10^3 \text{ J/kg}) = 6,664 \text{ W}$$

Heat transfer can also be expressed as

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s)$$

Then the required length of the tube becomes

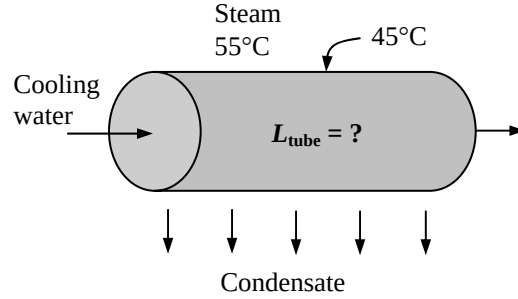
$$L = \frac{\dot{Q}}{h(\pi D)(T_{\text{sat}} - T_s)} = \frac{6664 \text{ W}}{(5644 \text{ W/m}^2\cdot^\circ\text{C})\pi(0.03 \text{ m})(55 - 45)^\circ\text{C}} = 1.21 \text{ m}$$

10-50 Saturated steam at a saturation temperature of $T_{\text{sat}} = 55^\circ\text{C}$ condenses on the outer surface of a horizontal tube which is maintained at 45°C . The required tube length to condense steam at a rate of 10 kg/h is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of water at the saturation temperature of 55°C are $h_{fg} = 2371 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.1045 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (55 + 45) / 2 = 50^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 988.1 \text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4181 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.644 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 2371 \times 10^3 \text{ J/kg} + 0.68 \times 4181 \text{ J/kg}\cdot^\circ\text{C}(55 - 45)^\circ\text{C} = 2399 \times 10^3 \text{ J/kg}\end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.8 \text{ m/s}^2)(988.1 \text{ kg/m}^3)(988.1 - 0.10 \text{ kg/m}^3)(2399 \times 10^3 \text{ J/kg})(0.644 \text{ W/m}\cdot^\circ\text{C})^3}{(0.547 \times 10^{-3} \text{ kg/m}\cdot\text{s})(55 - 45)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 10,135 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The rate of heat transfer during this condensation process is

$$\dot{Q} = \dot{m}h_{fg}^* = (10 / 3600 \text{ kg/s})(2399 \times 10^3 \text{ J/kg}) = 6,664 \text{ W}$$

Heat transfer can also be expressed as

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s)$$

Then the required length of the tube becomes

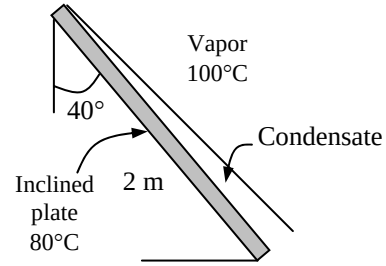
$$L = \frac{\dot{Q}}{h(\pi D)(T_{\text{sat}} - T_s)} = \frac{6664 \text{ W}}{(10,135 \text{ W/m}^2\cdot^\circ\text{C})\pi(0.03 \text{ m})(55 - 45)^\circ\text{C}} = \mathbf{0.70 \text{ m}}$$

10-51 Saturated steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a plate which is tilted 40° from the vertical and maintained at 80°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of the steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4206 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.675 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4206 \text{ J/kg}\cdot^\circ\text{C} (100 - 80)^\circ\text{C} = 2,314 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from the vertical plate relation by replacing g by $g \cos \theta$ where $\theta = 60^\circ$ to be

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{tilted, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.675 \text{ W/m}\cdot^\circ\text{C}) \times (100 - 80)^\circ\text{C}}{(0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s}) (2314 \times 10^3 \text{ J/kg})} \left(\frac{(9.8 \text{ m/s}^2) \cos 40}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 1197\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{tilted, wavy}} = \frac{\text{Re } k_l}{1.08 \text{ Re}^{1.22} - 5.2} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \\ &= \frac{1197 \times (0.675 \text{ W/m}\cdot^\circ\text{C})}{1.08 (1197)^{1.22} - 5.2} \left(\frac{(9.8 \text{ m/s}^2) \cos 40}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5438 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is: $A = w \times L = (2 \text{ m})(2 \text{ m}) = 4 \text{ m}^2$.

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA(T_{\text{sat}} - T_s) = (5438 \text{ W/m}^2 \cdot ^\circ\text{C})(4 \text{ m}^2)(100 - 80)^\circ\text{C} = 435,000 \text{ W}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{435,000 \text{ J/s}}{2314 \times 10^3 \text{ J/kg}} = 0.188 \text{ kg/s}$$

Discussion We could also determine the heat transfer coefficient from $h_{\text{inclined}} = h_{\text{vert}} (\cos \theta)^{1/4}$.

10-52 "PROBLEM 10-52"

"GIVEN"

T_sat=100 "[C]"

L=2 "[m]"

theta=40 "[degrees], parameter to be varied"

T_s=80 "[C], parameter to be varied"

"PROPERTIES"

Fluid\$='steam_NBS'

T_f=1/2*(T_sat+T_s)

P_sat=pressure(Fluid\$, T=T_sat, x=1)

rho_l=density(Fluid\$, T=T_f, x=0)

mu_l=Viscosity(Fluid\$, T=T_f, x=0)

nu_l=mu_l/rho_l

C_l=CP(Fluid\$, T=T_f, x=0)*Convert(kJ/kg-C, J/kg-C)

k_l=Conductivity(Fluid\$, T=T_f, P=P_sat+1)

h_f=enthalpy(Fluid\$, T=T_sat, x=0)

h_g=enthalpy(Fluid\$, T=T_sat, x=1)

h_fg=(h_g-h_f)*Convert(kJ/kg, J/kg)

g=9.8 "[m/s^2], gravitational acceleraton"

"ANALYSIS"

"(a)"

h_fg_star=h_fg+0.68*C_l*(T_sat-T_s)

Re=(4.81+(3.7*L*k_l*(T_sat-T_s))/(mu_l*h_fg_star))*((g*Cos(theta))/nu_l^2)^(1/3))^0.820

h=(Re*k_l)/(1.08*Re^1.22-5.2)*((g*Cos(theta))/nu_l^2)^(1/3)

Q_dot=h*A*(T_sat-T_s)

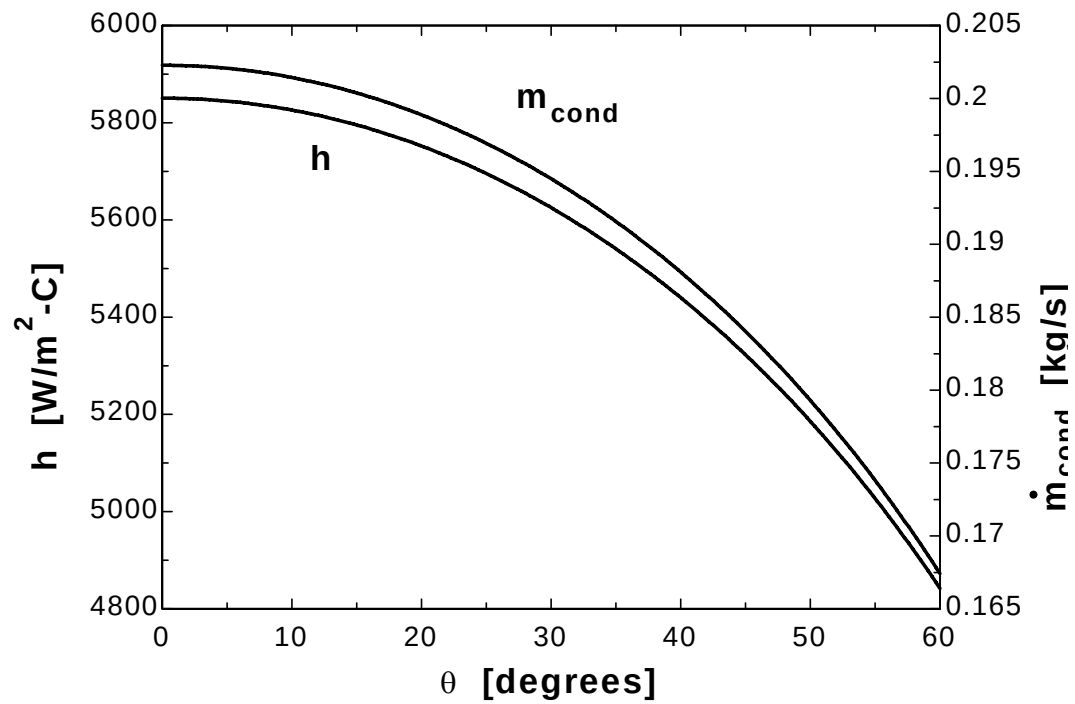
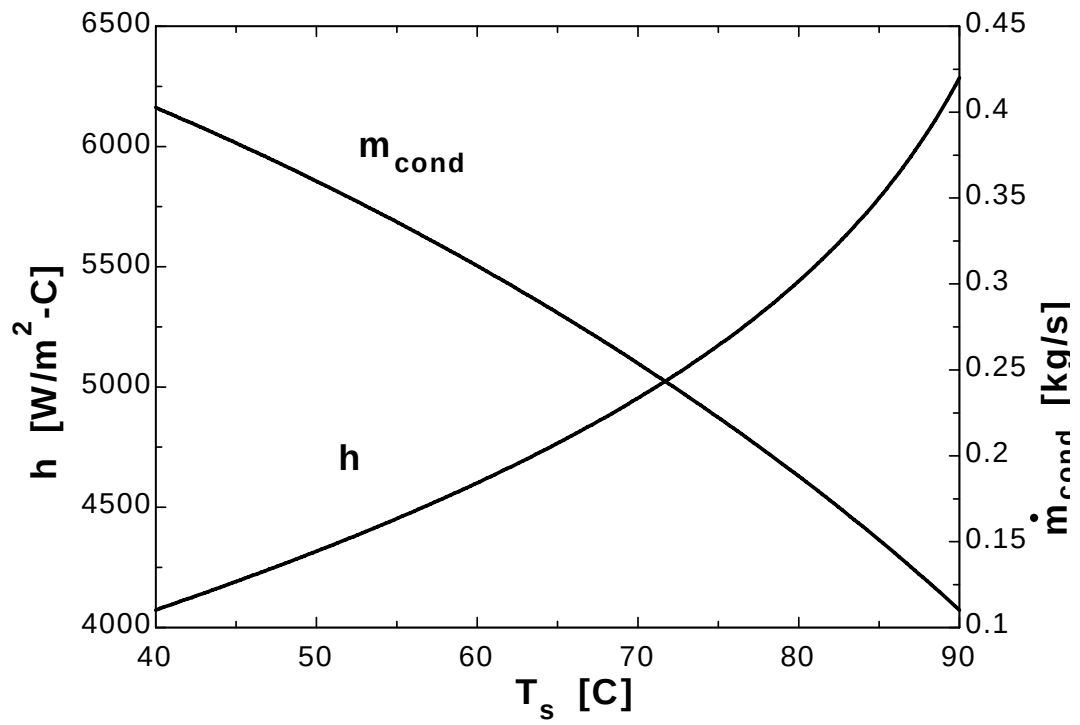
A=L^2

"(b)"

m_dot_cond=Q_dot/h_fg_star

T_s [C]	h [W/m ² .C]	m_{cond} [kg/s]
40	4073	0.4027
42.5	4131	0.3926
45	4191	0.3821
47.5	4253	0.3712
50	4317	0.3599
52.5	4383	0.3482
55	4453	0.3361
57.5	4525	0.3235
60	4601	0.3105
62.5	4681	0.2971
65	4766	0.2832
67.5	4857	0.2687
70	4954	0.2538
72.5	5059	0.2383
75	5173	0.2222
77.5	5299	0.2055
80	5440	0.1881
82.5	5600	0.1699
85	5786	0.151
87.5	6009	0.1311
90	6285	0.11

θ [degrees]	h [W/m ² .C]	m_{cond} [kg/s]
0	5851	0.2023
3	5848	0.2022
6	5842	0.202
9	5831	0.2016
12	5815	0.2011
15	5796	0.2004
18	5771	0.1995
21	5742	0.1985
24	5708	0.1974
27	5669	0.196
30	5625	0.1945
33	5576	0.1928
36	5522	0.1909
39	5462	0.1888
42	5395	0.1865
45	5323	0.184
48	5243	0.1813
51	5156	0.1783
54	5061	0.175
57	4956	0.1714
60	4842	0.1674

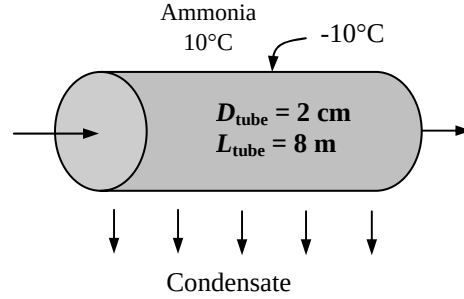


10-53 Saturated ammonia vapor at a saturation temperature of $T_{\text{sat}} = 10^\circ\text{C}$ condenses on the outer surface of a horizontal tube which is maintained at -10°C . The rate of heat transfer from the ammonia and the rate of condensation of ammonia are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of ammonia at the saturation temperature of 10°C are $h_{fg} = 1226 \times 10^3 \text{ J/kg}$ and $\rho_v = 4.870 \text{ kg/m}^3$ (Table A-11). The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (10 + (-10))/2 = 0^\circ\text{C}$ are (Table A-11),

$$\begin{aligned}\rho_l &= 638.6 \text{ kg/m}^3 \\ \mu_l &= 1.896 \times 10^{-4} \text{ kg/m} \cdot \text{s} \\ \nu_l &= 0.2969 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4617 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.5390 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1226 \times 10^3 \text{ J/kg} + 0.68 \times 4617 \text{ J/kg} \cdot ^\circ\text{C} [10 - (-10)]^\circ\text{C} = 1288 \times 10^3 \text{ J/kg}\end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(638.6 \text{ kg/m}^3)(638.6 - 4.870 \text{ kg/m}^3)(1288 \times 10^3 \text{ J/kg})(0.5390 \text{ W/m} \cdot ^\circ\text{C})^3}{(1.896 \times 10^{-4} \text{ kg/m} \cdot \text{s})[10 - (-10)]^\circ\text{C}(0.02 \text{ m})} \right]^{1/4} \\ &= 7390 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the tube is

$$A_s = \pi DL = \pi(0.02 \text{ m})(8 \text{ m}) = 0.5027 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (7390 \text{ W/m}^2 \cdot ^\circ\text{C})(0.5027 \text{ m}^2)[10 - (-10)]^\circ\text{C} = \mathbf{74,300 \text{ W}}$$

(b) The rate of condensation of ammonia is determined from

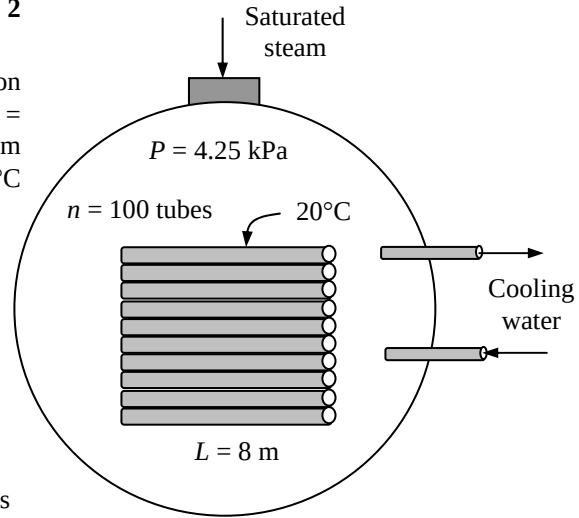
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{74,300 \text{ J/s}}{1288 \times 10^3 \text{ J/kg}} = \mathbf{0.0577 \text{ kg/s}}$$

10-54 Saturated steam at a pressure of 4.25 kPa and thus at a saturation temperature of $T_{\text{sat}} = 30^\circ\text{C}$ (Table A-9) condenses on the outer surfaces of 100 horizontal tubes arranged in a 10×10 square array maintained at 20°C by circulating cooling water. The rate of heat transfer to the cooling water and the rate of condensation of steam on the tubes are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 30°C are $h_{fg} = 2431 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.030 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (30 + 20) / 2 = 25^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 997.0 \text{ kg/m}^3 \\ \mu_l &= 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.894 \times 10^{-6} \text{ m}^2 / \text{s} \\ C_{pl} &= 4180 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.607 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$



Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2431 \times 10^3 \text{ J/kg} + 0.68 \times 4180 \text{ J/kg} \cdot ^\circ\text{C} (30 - 20)^\circ\text{C} = 2,459 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.8 \text{ m/s}^2)(997 \text{ kg/m}^3)(997 - 0.03 \text{ kg/m}^3)(2459 \times 10^3 \text{ J/kg})(0.607 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(30 - 20)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 8674 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 10-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{10^{1/4}} (8674 \text{ W/m}^2 \cdot ^\circ\text{C}) = 4878 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area for all 100 tubes is

$$A_s = N_{\text{total}} \pi D L = 100 \pi (0.03 \text{ m})(8 \text{ m}) = 75.40 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

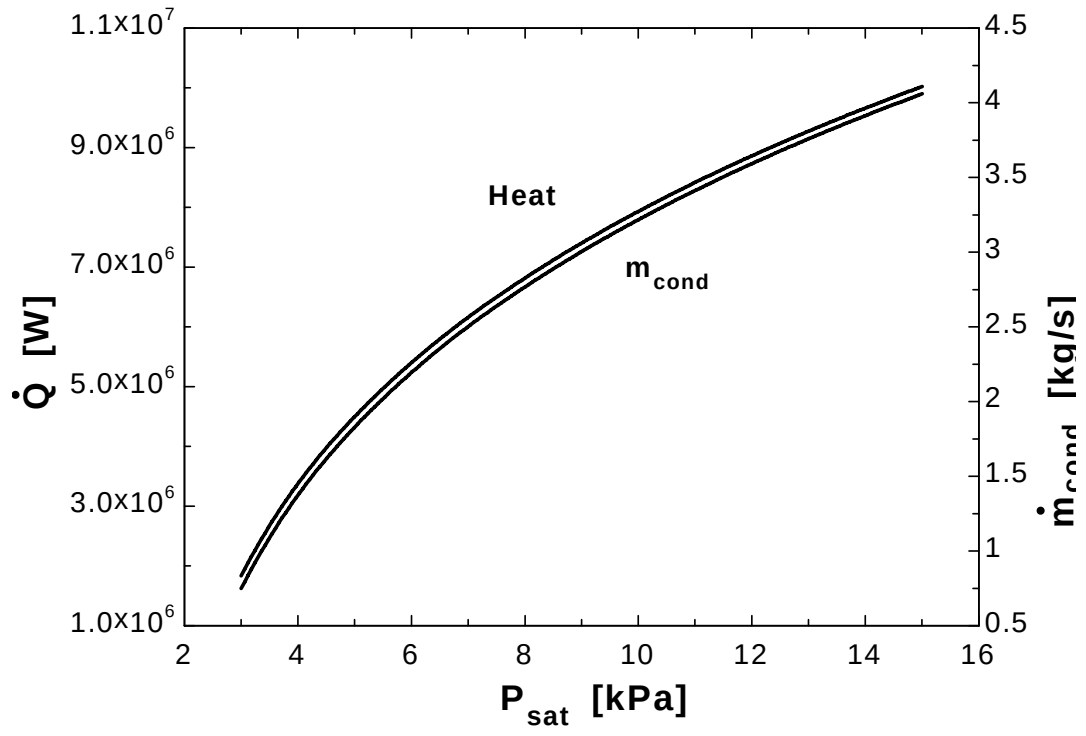
$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4878 \text{ W/m}^2 \cdot ^\circ\text{C})(75.40 \text{ m}^2)(30 - 20)^\circ\text{C} = 3,678,000 \text{ W} = \mathbf{3678 \text{ kW}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{3,678,000 \text{ J/s}}{2459 \times 10^3 \text{ J/kg}} = \mathbf{1.496 \text{ kg/s}}$$

10-55 "PROBLEM 10-55"**"GIVEN"****"P_{sat}=4.25 [kPa], parameter to be varied"****n_{tube}=100****N=10****L=8 "[m]"****D=0.03 "[m]"****T_s=20 "[C]"****"PROPERTIES"****Fluid\$='steam_NBS'****T_{sat}=temperature(Fluid\$, P=P_{sat}, x=1)****T_f=1/2*(T_{sat}+T_s)****h_f=enthalpy(Fluid\$, T=T_{sat}, x=0)****h_g=enthalpy(Fluid\$, T=T_{sat}, x=1)****h_{fg}=(h_g-h_f)*Convert(kJ/kg, J/kg)****rho_v=density(Fluid\$, T=T_{sat}, x=1)****rho_l=density(Fluid\$, T=T_f, x=0)****mu_l=Viscosity(Fluid\$, T=T_f, x=0)****nu_l=mu_l/rho_l****C_l=CP(Fluid\$, T=T_f, x=0)*Convert(kJ/kg-C, J/kg-C)****k_l=Conductivity(Fluid\$, T=T_f, P=P_{sat}+1)****g=9.8 "[m/s^2], gravitational acceleration"****"ANALYSIS"****"(a)"****h_{fg_star}=h_{fg}+0.68*C_l*(T_{sat}-T_s)****h_{1tube}=0.729*((g*rho_l*(rho_l-rho_v)*h_{fg_star}*k_l^3)/(mu_l*(T_{sat}-T_s*D)))^0.25****h=1/N^0.25*h_{1tube}****Q_{dot}=h*A*(T_{sat}-T_s)****A=n_{tube}*pi*D*L****"(b)"****m_{dot_cond}=Q_{dot}/h_{fg_star}**

P_{sat} [kPa]	Q [W]	m_{cond} [kg/s]
3	1836032	0.7478
4	3376191	1.374
5	4497504	1.829
6	5399116	2.194
7	6160091	2.502
8	6814744	2.766
9	7402573	3.004
10	7932545	3.218
11	8415994	3.413
12	8861173	3.592
13	9274152	3.758
14	9659732	3.914
15	10021650	4.059

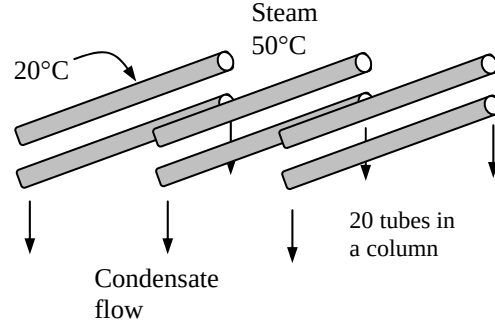


10-56 Saturated steam at a saturation temperature of $T_{\text{sat}} = 50^\circ\text{C}$ condenses on the outer surfaces of a tube bank with 20 tubes in each column maintained at 20°C . The average heat transfer coefficient and the rate of condensation of steam on the tubes are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 50°C are $h_{fg} = 2383 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.0831 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (50 + 20) / 2 = 35^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 994.0 \text{ kg/m}^3 \\ \mu_l &= 0.720 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.724 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4178 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.623 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 2383 \times 10^3 \text{ J/kg} + 0.68 \times 4178 \text{ J/kg}\cdot^\circ\text{C}(50 - 20)^\circ\text{C} \\ &= 2468 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.8 \text{ m/s}^2)(994 \text{ kg/m}^3)(994 - 0.08 \text{ kg/m}^3)(2468 \times 10^3 \text{ J/kg})(0.623 \text{ W/m}\cdot^\circ\text{C})^3}{(0.720 \times 10^{-3} \text{ kg/m}\cdot\text{s})(50 - 20)^\circ\text{C}(0.015 \text{ m})} \right]^{1/4} \\ &= 8425 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 10-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{20^{1/4}} (8425 \text{ W/m}^2\cdot^\circ\text{C}) = 3984 \text{ W/m}^2\cdot^\circ\text{C}$$

The surface area for all 20 tubes per unit length is

$$A_s = N_{\text{total}} \pi D L = 20 \pi (0.015 \text{ m})(1 \text{ m}) = 0.9425 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (3984 \text{ W/m}^2\cdot^\circ\text{C})(0.9425 \text{ m}^2)(50 - 20)^\circ\text{C} = 112,650 \text{ W}$$

(b) The rate of condensation of steam is determined from

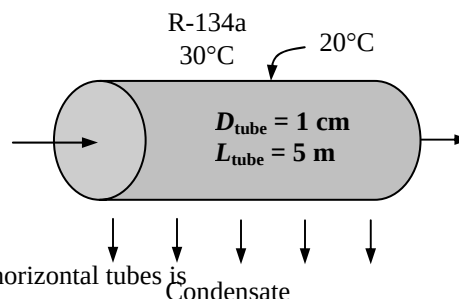
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{112,650 \text{ J/s}}{2468 \times 10^3 \text{ J/kg}} = 0.0456 \text{ kg/s}$$

10-57 Saturated refrigerant-134a vapor at a saturation temperature of $T_{\text{sat}} = 30^\circ\text{C}$ condenses inside a horizontal tube which is maintained at 20°C . The fraction of the refrigerant that will condense at the end of the tube is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vapor velocity is low so that $\text{Re}_{\text{vapor}} < 35,000$.

Properties The properties of refrigerant-134a at the saturation temperature of 30°C are $h_{fg} = 173.1 \times 10^3 \text{ J/kg}$ and $\rho_v = 37.53 \text{ kg/m}^3$. The properties of liquid R-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (30 + 20) / 2 = 25^\circ\text{C}$ are (Table A-10)

$$\begin{aligned}\rho_l &= 1207 \text{ kg/m}^3 \\ \mu_l &= 2.012 \times 10^{-4} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.1667 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 1427 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.08325 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The heat transfer coefficient for condensation inside horizontal tubes is

$$\begin{aligned}h &= h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s)} \left(h_{fg} + \frac{3}{8} C_{pl} (T_{\text{sat}} - T_s) \right) \right]^{1/4} \\ &= 0.555 \left[\frac{(9.81 \text{ m/s}^2)(1207 \text{ kg/m}^3)(1207 - 37.53) \text{ kg/m}^3 (0.08325 \text{ W/m}\cdot^\circ\text{C})^3}{(2.012 \times 10^{-4} \text{ kg/m}\cdot\text{s})(30 - 20)^\circ\text{C}} \right. \\ &\quad \left. \times \left(173.1 \times 10^3 \text{ J/kg} + \frac{3}{8} (1427 \text{ J/kg}\cdot^\circ\text{C})(30 - 20)^\circ\text{C} \right) \right]^{1/4} \\ &= 509.2 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the pipe is

$$A_s = \pi D L = \pi (0.01 \text{ m})(5 \text{ m}) = 0.1571 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (509.2 \text{ W/m}^2 \cdot ^\circ\text{C})(0.1571 \text{ m}^2)(30 - 20)^\circ\text{C} = 800.0 \text{ W}$$

The modified latent heat of vaporization in this case is, as indicated in the h relation,

$$h_{fg}^* = h_{fg} + \frac{3}{8} C_{pl} (T_{\text{sat}} - T_s) = 173.1 \times 10^3 \text{ J/kg} + \frac{3}{8} (1427 \text{ J/kg}\cdot^\circ\text{C})(30 - 20)^\circ\text{C} = 178.5 \times 10^3 \text{ J/kg}$$

Then the rate of condensation becomes

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{800 \text{ J/s}}{178.5 \times 10^3 \text{ J/kg}} = 0.004482 \text{ kg/s} = 0.2689 \text{ kg/min}$$

Therefore, the fraction of the refrigerant that will condense at the end of the tube is

$$\text{Fraction condensed} = \frac{\dot{m}_{\text{condensed}}}{\dot{m}_{\text{total}}} = \frac{0.2689 \text{ kg/min}}{2.5 \text{ kg/min}} = \mathbf{0.108 \text{ (or 10.8\%)}}$$

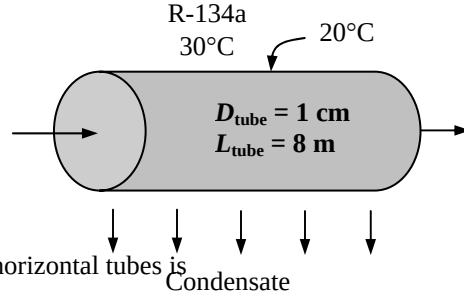
Discussions Note that we used the modified h_{fg}^* instead of just h_{fg} in heat transfer calculations to account for heat transfer due to the cooling of the condensate below the saturation temperature.

10-58 Saturated refrigerant-134a vapor condenses inside a horizontal tube maintained at a uniform temperature. The fraction of the refrigerant that will condense at the end of the tube is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vapor velocity is low so that $Re_{\text{vapor}} < 35,000$.

Properties The properties of refrigerant-134a at the saturation temperature of 30°C are $h_{fg} = 173.1 \times 10^3 \text{ J/kg}$ and $\rho_v = 37.53 \text{ kg/m}^3$. The properties of liquid R-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (30 + 20) / 2 = 25^\circ\text{C}$ are (Table A-10)

$$\begin{aligned}\rho_l &= 1207 \text{ kg/m}^3 \\ \mu_l &= 2.012 \times 10^{-4} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.1667 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 1427 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.08325 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The heat transfer coefficient for condensation inside horizontal tubes is

$$\begin{aligned}h &= h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s)} \left(h_{fg} + \frac{3}{8} C_{pl} (T_{\text{sat}} - T_s) \right) \right]^{1/4} \\ &= 0.555 \left[\frac{(9.81 \text{ m/s}^2)(1207 \text{ kg/m}^3)(1207 - 37.53) \text{ kg/m}^3 (0.08325 \text{ W/m}\cdot^\circ\text{C})^3}{(2.012 \times 10^{-4} \text{ kg/m}\cdot\text{s})(30 - 20)^\circ\text{C}} \right. \\ &\quad \left. \times \left(173.1 \times 10^3 \text{ J/kg} + \frac{3}{8} (1427 \text{ J/kg}\cdot^\circ\text{C})(30 - 20)^\circ\text{C} \right) \right]^{1/4} \\ &= 509.2 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the pipe is

$$A_s = \pi D L = \pi (0.01 \text{ m})(8 \text{ m}) = 0.2513 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (509.2 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2513 \text{ m}^2)(30 - 20)^\circ\text{C} = 1280 \text{ W}$$

The modified latent heat of vaporization in this case is, as indicated in the h relation,

$$h_{fg}^* = h_{fg} + \frac{3}{8} C_{pl} (T_{\text{sat}} - T_s) = 173.1 \times 10^3 \text{ J/kg} + \frac{3}{8} (1427 \text{ J/kg}\cdot^\circ\text{C})(30 - 20)^\circ\text{C} = 178.5 \times 10^3 \text{ J/kg}$$

Then the rate of condensation becomes

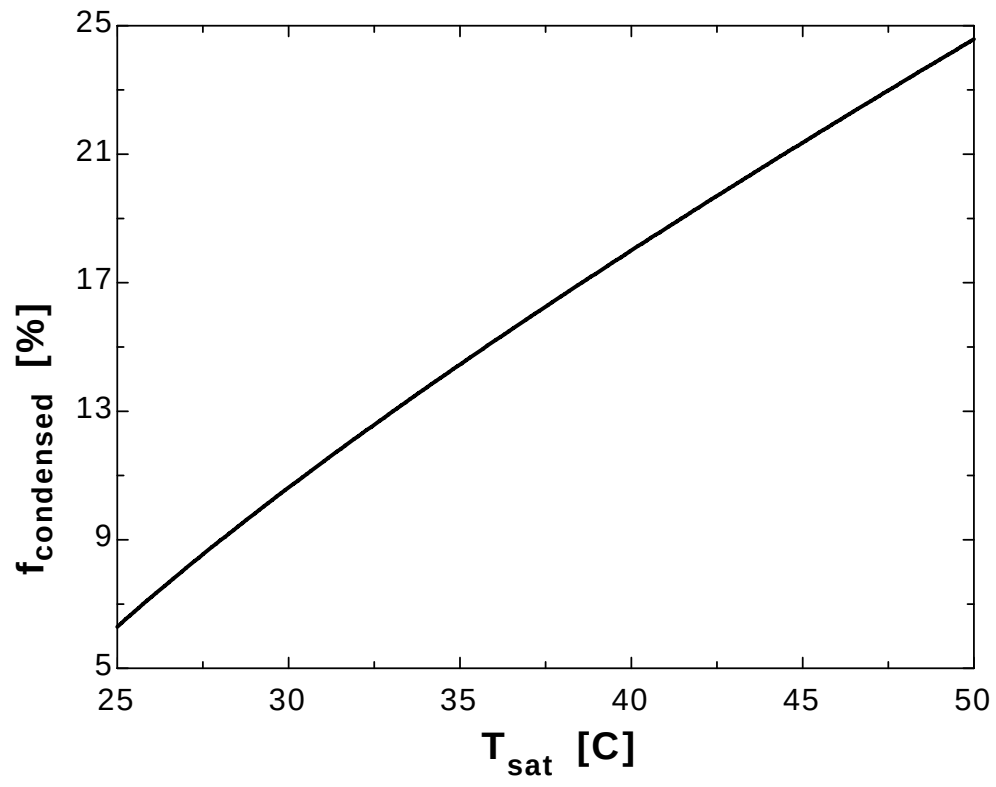
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1280 \text{ J/s}}{178.5 \times 10^3 \text{ J/kg}} = 0.007169 \text{ kg/s} = 0.4301 \text{ kg/min}$$

Therefore, the fraction of the refrigerant that will condense at the end of the tube is

$$\text{Fraction condensed} = \frac{\dot{m}_{\text{condensed}}}{\dot{m}_{\text{total}}} = \frac{0.4301 \text{ kg/min}}{2.5 \text{ kg/min}} = \mathbf{0.172 \text{ (or 17.2%)}}$$

10-59 "PROBLEM 10-59"**"GIVEN"****"T_{sat}=30 [C], parameter to be varied"****L=5 "[m]"****D=0.01 "[m]"****T_s=20 "[C]"****m_{dot}total=2.5 "[kg/min]"****"PROPERTIES"****rho_l=1187 "[kg/m^3]"****rho_v=37.5 "[kg/m^3]"****mu_l=0.201E-3 "[kg/m-s]"****C_l=1447 "[J/kg-C]"****k_l=0.0796 "[W/m-C]"****h_{fg}=173.3E3 "[J/kg]"****g=9.8 "[m/s^2], gravitational acceleration"****"ANALYSIS"****h=0.555*((g*rho_l*(rho_l-rho_v)*k_l^3)/(mu_l*(T_{sat}-T_s))*(h_{fg}+3/8*C_l*(T_{sat}-T_s)))^0.25****Q_{dot}=h*A*(T_{sat}-T_s)****A=pi*D*L****m_{dot}cond=Q_{dot}/h_{fg}*Convert(kg/s, kg/min)****f_{condensed}=m_{dot}cond/m_{dot}total*Convert(,%)**

T_{sat} [C]	f_{condensed} [%]
25	6.293
26.25	7.447
27.5	8.546
28.75	9.603
30	10.62
31.25	11.62
32.5	12.58
33.75	13.53
35	14.45
36.25	15.36
37.5	16.26
38.75	17.14
40	18
41.25	18.86
42.5	19.7
43.75	20.53
45	21.36
46.25	22.18
47.5	22.98
48.75	23.78
50	24.58



Special Topic: Heat Pipes

10-60C A heat pipe is a simple device with no moving parts which can transfer large quantities of heat over fairly large distances essentially at a constant temperature without requiring any power input. A heat pipe is basically a sealed slender tube containing a wick structure lined on the inner surface and a small amount of fluid such as water at the saturated state. It is composed of three sections: the evaporator section at one end where heat is absorbed and the fluid is vaporized, a condenser section at the other end where the vapor is condensed and heat is rejected, and the adiabatic section in between where the vapor and the liquid phases of the fluid flow in opposite directions through the core and the wick, respectively, to complete the cycle with no significant heat transfer between the fluid and the surrounding medium.

10-61C The boiling and condensation processes are associated with extremely high heat transfer coefficients, and thus it is natural to expect the heat pipe to be an extremely effective heat transfer device since its operation is based on alternate boiling and condensation of the working fluid.

10-62C The non-condensable gases such as air degrade the performance of the heat pipe, and can destroy it in a short time.

10-63C The heat pipes with water as the working fluid are designed to remove heat at temperatures below the boiling temperature of water at atmospheric pressure (i.e., 100°C). Therefore, the pressure inside the heat pipe must be maintained below the atmospheric pressure to provide boiling at such temperatures.

10-64C Liquid motion in the wick depends on the dynamic balance between two opposing effects: the capillary pressure which pumps the liquid through the pores, and the internal resistance to flow due to the friction between the mesh surface and the liquid. Small pores increase the capillary action, but it also increases the friction force opposing the fluid motion. At optimum core size, the difference between the capillary force and the friction force is maximum.

10-65C The orientation of the heat pipe affects its performance. The performance of a heat pipe will be best when the capillary and gravity forces act in the same direction (evaporator end down), and it will be worst when these two forces act in opposite directions (evaporator end up).

10-66C The capillary pressure which creates the suction effect to draw the liquid forces the liquid in a heat pipe to move up against the gravity without a pump. For the heat pipes which work against the gravity, it is better to have fine wicks since the capillary pressure is inversely proportional to the effective capillary radius of the mesh.

10-67C The most important consideration in the design of a heat pipe is the compatibility of the materials used for the tube, wick and the fluid.

10-68C The major cause for the premature degradation of the performance of some heat pipes is contamination which occurs during the sealing of the ends of the heat pipe tube.

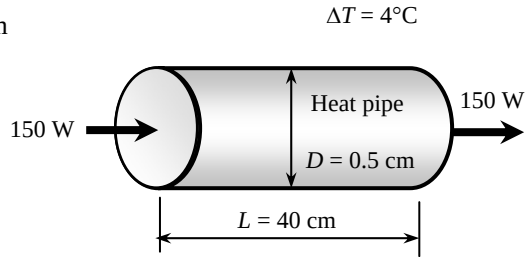
10-69 A 40-cm long cylindrical heat pipe dissipates heat at a rate of 150 W. The diameter and mass of a 40-cm long copper rod that can conduct heat at the same rate are to be determined.

Assumptions Steady operating conditions exist.

Properties The properties of copper at room temperature are $\rho = 8950 \text{ kg/m}^3$ and $k = 386 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The rate of heat transfer through the copper rod can be expressed as

$$\dot{Q} = kA \frac{\Delta T}{L}$$



Solving for the cross-sectional area A and the diameter D gives

$$A = \frac{L}{k\Delta T} \dot{Q} = \frac{0.4 \text{ m}}{(386 \text{ W/m}\cdot^\circ\text{C})(4^\circ\text{C})} (150 \text{ W}) = 0.03886 \text{ m}^2 = 388.6 \text{ cm}^2$$

$$A = \frac{\pi D^2}{4} \longrightarrow D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(388.6 \text{ cm}^2)}{\pi}} = \mathbf{22.2 \text{ cm}}$$

The mass of this copper rod is

$$m = \rho V = \rho AL = (8950 \text{ kg/m}^3)(0.03886 \text{ m}^2)(0.4 \text{ m}) = \mathbf{139 \text{ kg}}$$

10-70 A 40-cm long cylindrical heat pipe dissipates heat at a rate of 150 W. The diameter and mass of a 40-cm long copper rod that can conduct heat at the same rate are to be determined.

Assumptions Steady operating conditions exist.

Properties The properties of copper at room temperature are $\rho = 2702 \text{ kg/m}^3$ and $k = 237 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The rate of heat transfer through the aluminum rod can be expressed as

$$\dot{Q} = kA \frac{\Delta T}{L}$$

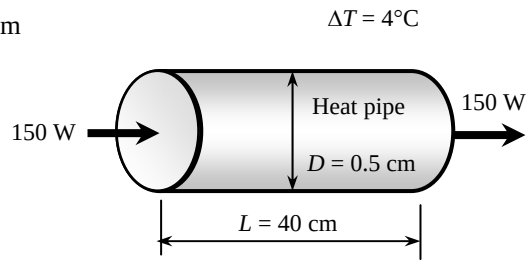
Solving for the cross-sectional area A and the diameter D gives

$$A = \frac{L}{k\Delta T} \dot{Q} = \frac{0.4 \text{ m}}{(237 \text{ W/m}\cdot^\circ\text{C})(4^\circ\text{C})} (150 \text{ W}) = 0.06329 \text{ m}^2 = 632.9 \text{ cm}^2$$

$$A = \frac{\pi D^2}{4} \longrightarrow D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(632.9 \text{ cm}^2)}{\pi}} = \mathbf{28.4 \text{ cm}}$$

The mass of this aluminum rod is

$$m = \rho V = \rho AL = (2702 \text{ kg/m}^3)(0.06329 \text{ m}^2)(0.4 \text{ m}) = \mathbf{68.4 \text{ kg}}$$



10-71E A plate that supports 10 power transistors is to be cooled with heat pipes. The number of pipes need to be attached to this plate is to be determined.

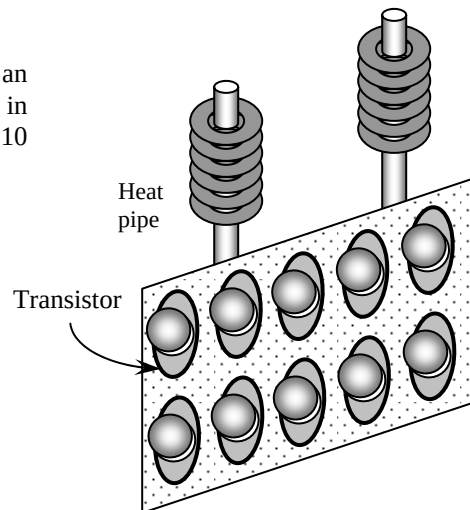
Assumptions Steady operating conditions exist.

Analysis The heat removal rate of heat pipes that have an outside diameter of 0.635 cm and length of 30.5 cm is given in Table 10-5 to be 175 W. The total rate of heat dissipated by 10 transistors each dissipating 35 W is

$$\dot{Q}_{total} = (10)(35 \text{ W}) = 350 \text{ W}$$

Then the number of heat pipes need to be attached to the plate becomes

$$n = \frac{\dot{Q}_{total}}{\dot{Q}} = \frac{350 \text{ W}}{175 \text{ W}} = \mathbf{2}$$



Review Problems

10-72 Steam at a saturation temperature of $T_{\text{sat}} = 40^\circ\text{C}$ condenses on the outside of a thin horizontal tube. Heat is transferred to the cooling water that enters the tube at 25°C and exits at 35°C . The rate of condensation of steam, the average overall heat transfer coefficient, and the tube length are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube can be taken to be isothermal at the bulk mean fluid temperature in the evaluation of the condensation heat transfer coefficient. 3 Liquid flow through the tube is fully developed. 4 The thickness and the thermal resistance of the tube is negligible.

Properties The properties of water at the saturation temperature of 40°C are $h_{\text{fg}} = 2407 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.05 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (50 + 20)/2 = 35^\circ\text{C}$ and at the bulk fluid temperature of $T_b = (T_{\text{in}} + T_{\text{out}})/2 = (25 + 35)/2 = 30^\circ\text{C}$ are (Table A-9),

At 35°C :

$$\rho_l = 994.0 \text{ kg/m}^3$$

$$\mu_l = 0.720 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

$$C_{pl} = 4178 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.623 \text{ W/m}\cdot^\circ\text{C}$$

At 30°C :

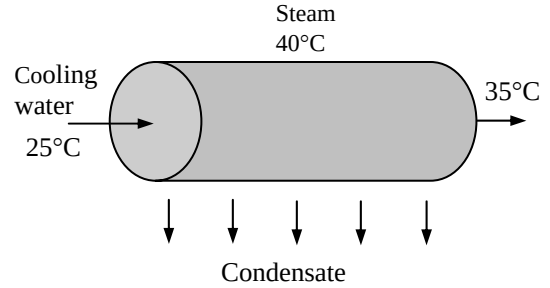
$$\rho_l = 996.0 \text{ kg/m}^3$$

$$\nu_l = \mu_l / \rho_l = 0.801 \times 10^{-6} \text{ m}^2/\text{s}$$

$$C_{pl} = 4178 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.615 \text{ W/m}\cdot^\circ\text{C}$$

$$\text{Pr} = 5.42$$



Analysis The mass flow rate of water and the rate of heat transfer to the water are

$$\dot{m}_{\text{water}} = \rho V A_c = (996 \text{ kg/m}^3)(2 \text{ m/s})[\pi(0.03 \text{ m})^2/4] = 1.408 \text{ kg/s}$$

$$\dot{Q} = \dot{m} C_p (T_{\text{out}} - T_{\text{in}}) = (1.408 \text{ kg/s})(4178 \text{ J/kg}\cdot^\circ\text{C})(35 - 25)^\circ\text{C} = \mathbf{58,820 \text{ W}}$$

The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2407 \times 10^3 \text{ J/kg} + 0.68 \times 4178 \text{ J/kg}\cdot^\circ\text{C} (40 - 30)^\circ\text{C} = 2435 \times 10^3 \text{ J/kg} \end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned} h_o &= h_{\text{horizontal}} = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.8 \text{ m/s}^2)(994 \text{ kg/m}^3)(994 - 0.05 \text{ kg/m}^3)(2435 \times 10^3 \text{ J/kg})(0.623 \text{ W/m}\cdot^\circ\text{C})^3}{(0.720 \times 10^{-3} \text{ kg/m}\cdot\text{s})(40 - 30)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 9292 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The average heat transfer coefficient for flow inside the tube is determined as follows:

$$\begin{aligned} \text{Re} &= \frac{V_m D}{\nu} = \frac{(2 \text{ m/s})(0.03 \text{ m})}{0.801 \times 10^{-6}} = 74,906 \\ \text{Nu} &= 0.023 \text{Re}^{0.8} \text{Pr}^{0.4} = 0.023(74,906)^{0.8} (5.42)^{0.4} = 359 \\ h_i &= \frac{k \text{Nu}}{D} = \frac{(0.615 \text{ W/m}\cdot^\circ\text{C}) \times 359}{0.03 \text{ m}} = 7357 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

Noting that the thermal resistance of the tube is negligible, the overall heat transfer coefficient becomes

$$U = \frac{1}{1/h_i + 1/h_o} = \frac{1}{1/7357 + 1/9292} = \mathbf{4106 \text{ W/m}^2\cdot^\circ\text{C}}$$

The logarithmic mean temperature difference is:
$$\Delta T_{\text{ln}} = \frac{\Delta T_i - \Delta T_e}{\ln(\Delta T_i / \Delta T_e)} = \frac{15 - 5}{\ln(15/5)} = 9.10^\circ\text{C}$$

The tube length is determined from

$$\dot{Q} = hA_s \Delta T_{\text{ln}} \rightarrow L = \frac{\dot{Q}}{h(\pi D) \Delta T_{\text{ln}}} = \frac{58,820 \text{ W}}{(4106 \text{ W/m}^2 \cdot ^\circ\text{C}) \pi (0.03 \text{ m}) (9.10^\circ\text{C})} = \mathbf{16.7 \text{ m}}$$

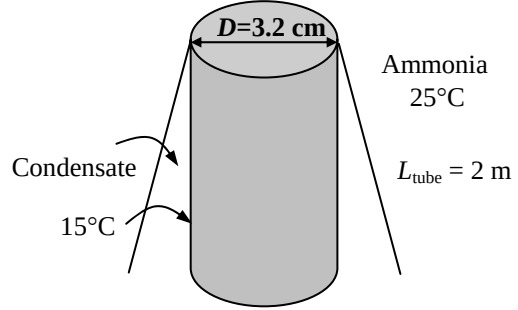
Note that the flow is turbulent, and thus the entry length in this case is $10D = 0.3 \text{ m}$ is much shorter than the total tube length. This verifies our assumption of fully developed flow.

10-73 Saturated ammonia at a saturation temperature of $T_{\text{sat}} = 25^\circ\text{C}$ condenses on the outer surface of vertical tube which is maintained at 15°C by circulating cooling water. The rate of heat transfer to the coolant and the rate of condensation of ammonia are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The tube can be treated as a vertical plate. 4 The condensate flow is turbulent over the entire tube (this assumption will be verified). 5 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of ammonia at the saturation temperature of 25°C are $h_{fg} = 1166 \times 10^3 \text{ J/kg}$ and $\rho_v = 7.809 \text{ kg/m}^3$. The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (25 + 15) / 2 = 20^\circ\text{C}$ are (Table A-11),

$$\begin{aligned}\rho_l &= 610.2 \text{ kg/m}^3 \\ \mu_l &= 1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.2489 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 4745 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.4927 \text{ W/m} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.463\end{aligned}$$



Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 1166 \times 10^3 \text{ J/kg} + 0.68 \times 4745 \text{ J/kg} \cdot ^\circ\text{C} (25 - 15)^\circ\text{C} = 1198 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming turbulent flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vertical, turb}} = \frac{\text{Re } k_l}{8750 + 58 \text{Pr}^{-0.5} (\text{Re}^{0.75} - 253) \mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{2149 \times (0.4827 \text{ W/m} \cdot ^\circ\text{C})}{8750 + 58 \times 1.463^{-0.5} (2149^{0.75} - 253) (1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s}) (1198 \times 10^3 \text{ J/kg})} \left(\frac{9.81 \text{ m/s}^2}{(0.2489 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \\ &= 2140\end{aligned}$$

which is greater than 1800, and thus our assumption of turbulent flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{vertical, turbulent}} = \frac{\text{Re } k_l}{8750 + 58 \text{Pr}^{-0.5} (\text{Re}^{0.75} - 253)} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{2149 \times (0.4827 \text{ W/m} \cdot ^\circ\text{C})}{8750 + 58 \times 1.463^{-0.5} (2149^{0.75} - 253)} \left(\frac{9.81 \text{ m/s}^2}{(0.2489 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 4871 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the tube is $A_s = \pi DL = \pi(0.032 \text{ m})(2 \text{ m}) = 0.2011 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4871 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2011 \text{ m}^2)(25 - 15)^\circ\text{C} = \mathbf{9794 \text{ W}}$$

(b) The rate of condensation of ammonia is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{9794 \text{ J/s}}{1198 \times 10^3 \text{ J/kg}} = \mathbf{8.175 \times 10^{-3} \text{ kg/s}}$$

10-74 There is film condensation on the outer surfaces of 8 horizontal tubes arranged in a horizontal or vertical tier. The ratio of the condensation rate for the cases of the tubes being arranged in a horizontal tier versus in a vertical tier is to be determined.

Assumptions Steady operating conditions exist.

Analysis The heat transfer coefficients for the two cases are related to the heat transfer coefficient on a single horizontal tube by

Horizontal tier: $h_{\text{horizontal tier of } N \text{ tubes}} = h_{\text{horizontal, 1 tube}}$

Vertical tier: $h_{\text{vertical tier of } N \text{ tubes}} = \frac{h_{\text{horizontal, 1 tube}}}{N^{1/4}}$



Horizontal tier



Vertical tier

Therefore,

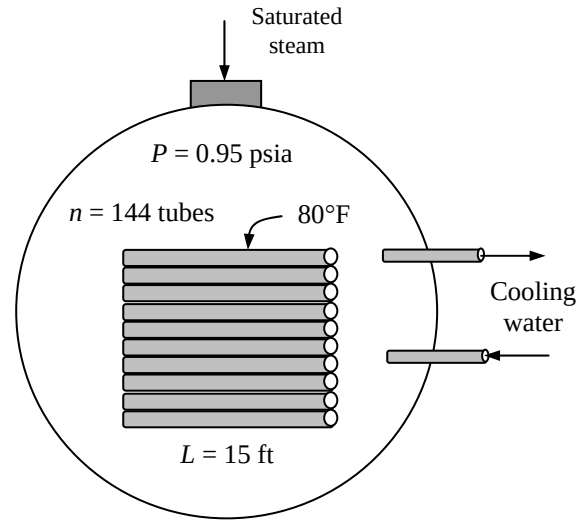
$$\begin{aligned} \text{Ratio} &= \frac{\dot{m}_{\text{horizontal tier of } N \text{ tubes}}}{\dot{m}_{\text{vertical tier of } N \text{ tubes}}} \\ &= \frac{h_{\text{horizontal tier of } N \text{ tubes}}}{h_{\text{vertical tier of } N \text{ tubes}}} \\ &= \frac{h_{\text{horizontal, 1 tube}}}{h_{\text{horizontal, 1 tube}} / N^{1/4}} \\ &= N^{1/4} \\ &= 8^{1/4} = \mathbf{1.68} \end{aligned}$$

10-75E Saturated steam at a saturation pressure of 0.95 psia and thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{F}$ (Table A-9E) condenses on the outer surfaces of 144 horizontal tubes which are maintained at 80°F by circulating cooling water and arranged in a 12×12 square array. The rate of heat transfer to the cooling water and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 100°F are $h_{fg} = 1037$ Btu/lbm and $\rho_v = 0.00286$ lbm/ft³. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{F}$ are (Table A-9E),

$$\begin{aligned}\rho_l &= 62.12 \text{ lbm/ft}^3 \\ \mu_l &= 1.842 \text{ lbm/ft} \cdot \text{h} \\ \nu_l &= \mu_l / \rho_l = 0.02965 \text{ ft}^2 / \text{h} \\ C_{pl} &= 0.999 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_l &= 0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1037 \text{ Btu/lbm} + 0.68 \times (0.999 \text{ Btu/lbm} \cdot ^\circ\text{F})(100 - 80)^\circ\text{F} \\ &= 1051 \text{ Btu/lbm}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(32.2 \text{ ft/s}^2)(62.12 \text{ lbm/ft}^3)(62.12 - 0.00286 \text{ lbm/ft}^3)(1051 \text{ Btu/lbm})(0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F})^3}{[(1 \text{ h}/3600 \text{ s})^2](1.842 \text{ lbm/ft} \cdot \text{h})(100 - 80)^\circ\text{F}(1.2/12 \text{ ft})} \right]^{1/4} \\ &= 1562 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}\end{aligned}$$

Then the average heat transfer coefficient for a 4-tube high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{12^{1/4}} (1562 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) = 839 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$$

The surface area for all 144 tubes is

$$A_s = N_{\text{total}} \pi D L = 144 \pi (1.2/12 \text{ ft})(15 \text{ ft}) = 678.6 \text{ ft}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (839 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(678.6 \text{ ft}^2)(100 - 80)^\circ\text{F} = \mathbf{11,387,000 \text{ Btu/h}}$$

(b) The rate of condensation of steam is determined from

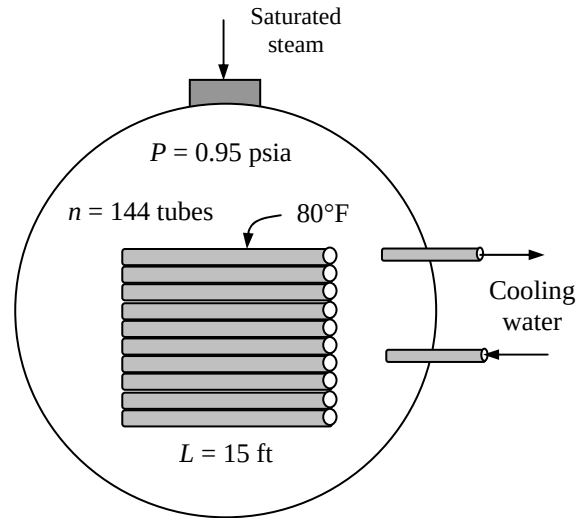
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{11,387,000 \text{ Btu/h}}{1051 \text{ Btu/lbm}} = \mathbf{10,830 \text{ lbm/h}}$$

10-76E Saturated steam at a saturation pressure of 0.95 psia and thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{F}$ (Table A-9E) condenses on the outer surfaces of 144 horizontal tubes which are maintained at 80°F by circulating cooling water and arranged in a 12×12 square array. The rate of heat transfer to the cooling water and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 10°F are $h_{fg} = 1037$ Btu/lbm and $\rho_v = 0.00286$ lbm/ft³. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{F}$ are (Table A-9E),

$$\begin{aligned}\rho_l &= 62.12 \text{ lbm/ft}^3 \\ \mu_l &= 1.842 \text{ lbm/ft} \cdot \text{h} \\ \nu_l &= \mu_l / \rho_l = 0.02965 \text{ ft}^2 / \text{h} \\ C_{pl} &= 0.999 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_l &= 0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$



Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1037 \text{ Btu/lbm} + 0.68 \times (0.999 \text{ Btu/lbm} \cdot ^\circ\text{F})(100 - 80)^\circ\text{F} \\ &= 1051 \text{ Btu/lbm}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(32.2 \text{ ft/s}^2)(62.12 \text{ lbm/ft}^3)(62.12 - 0.00286 \text{ lbm/ft}^3)(1051 \text{ Btu/lbm})(0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F})^3}{[(1 \text{ h}/3600 \text{ s})^2](1.842 \text{ lbm/ft} \cdot \text{h})(100 - 80)^\circ\text{F}(2.0/12 \text{ ft})} \right]^{1/4} \\ &= 1374 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}\end{aligned}$$

Then the average heat transfer coefficient for a 4-tube high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{12^{1/4}} (1374 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) = 739 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$$

The surface area for all 144 tubes is

$$A_s = N_{\text{total}} \pi D L = 144 \pi (2/12 \text{ ft})(15 \text{ ft}) = 1131 \text{ ft}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (739 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(1131 \text{ ft}^2)(100 - 80)^\circ\text{F} = \mathbf{16,716,000 \text{ Btu/h}}$$

(b) The rate of condensation of steam is determined from

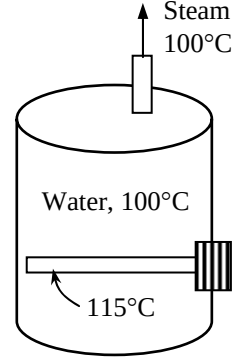
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{16,716,000 \text{ Btu/h}}{1051 \text{ Btu/lbm}} = \mathbf{15,900 \text{ lbm/h}}$$

10-77 Water is boiled at $T_{\text{sat}} = 100^\circ\text{C}$ by a chemically etched stainless steel electric heater whose surface temperature is maintained at $T_s = 115^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation of water, and the maximum rate of evaporation are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg}\cdot\text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg}\cdot^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a chemically etched stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 115 - 100 = 15^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(115 - 100)}{(0.0130)(2257 \times 10^3)1.75} \right)^3 \\ &= 474,900 \text{ W/m}^2\end{aligned}$$

The surface area of the bottom of the heater is $A_s = \pi DL = \pi(0.002 \text{ m})(0.8 \text{ m}) = 0.005027 \text{ m}^2$.

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.005027 \text{ m}^2)(474,900 \text{ W/m}^2) = \mathbf{2387 \text{ W}}$$

The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{2387 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{1.058 \times 10^{-3} \text{ kg/s} = 3.81 \text{ kg/h}}$$

(b) For a horizontal heating wire, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.001) \left(\frac{9.8(957.9 - 0.60)}{0.0589} \right)^{1/2} = 0.399 < 1.2 \\ C_{cr} &= 0.12 L^{*-0.25} = 0.12(0.399)^{-0.25} = 0.151\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} = 0.151(2257 \times 10^3)[0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= \mathbf{1,280,000 \text{ W/m}^2}\end{aligned}$$

10-78E Steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{F}$ condenses on a vertical plate which is maintained at 80°C . The rate of heat transfer to the plate and the rate of condensation of steam per ft width of the plate are to be determined.

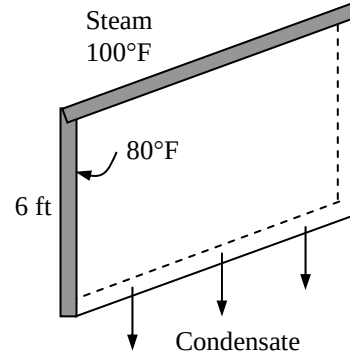
Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°F are $h_{fg} = 1037 \text{ Btu/lbm}$ and $\rho_v = 0.00286 \text{ lbm/ft}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{F}$ are (Table A-9E),

$$\begin{aligned}\rho_l &= 62.12 \text{ lbm/ft}^3 \\ \mu_l &= 1.842 \text{ lbm/ft} \cdot \text{h} \\ \nu_l &= \mu_l / \rho_l = 0.02965 \text{ ft}^2 / \text{h} \\ C_{pl} &= 0.999 \text{ Btu/lbm} \cdot ^\circ\text{F} \\ k_l &= 0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 1037 \text{ Btu/lbm} + 0.68 \times (0.999 \text{ Btu/lbm} \cdot ^\circ\text{F})(100 - 80)^\circ\text{F} \\ &= 1051 \text{ Btu/lbm}\end{aligned}$$



Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70Lk_l(T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (6 \text{ ft}) \times (0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \times (100 - 80)^\circ\text{F}}{(1.842 \text{ lbm/ft} \cdot \text{h})(1051 \text{ Btu/lbm})} \left(\frac{32.2 \text{ ft/s}^2}{(0.02965 \text{ ft}^2 / \text{h})^2} \frac{(3600 \text{ s})^2}{(1 \text{ h})^2} \right)^{1/3} \right]^{0.82} = 201\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re} k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{201 \times (0.358 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F})}{1.08(201)^{1.22} - 5.2} \left(\frac{32.2 \text{ ft/s}^2}{(0.02965 \text{ ft}^2 / \text{h})^2} \frac{(3600 \text{ s})^2}{(1 \text{ h})^2} \right)^{1/3} = 813 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}\end{aligned}$$

The heat transfer surface area of the plate is

$$A_s = W \times L = (6 \text{ ft})(1 \text{ ft}) = 6 \text{ ft}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (813 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(6 \text{ ft}^2)(100 - 80)^\circ\text{F} = \mathbf{97,530 \text{ Btu/h}}$$

The rate of condensation of steam is determined from

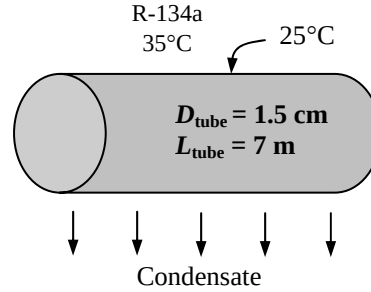
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{97,530 \text{ Btu/h}}{1051 \text{ Btu/lbm}} = \mathbf{92.8 \text{ lbm/h}}$$

10-79 Saturated refrigerant-134a vapor condenses on the outside of a horizontal tube maintained at a specified temperature. The rate of condensation of the refrigerant is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of refrigerant-134a at the saturation temperature of 35°C are $h_{fg} = 168.2 \times 10^3$ J/kg and $\rho_v = 43.41$ kg/m³. The properties of liquid R-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (35 + 25) / 2 = 30^\circ\text{C}$ are (Table A-10),

$$\begin{aligned}\rho_l &= 1188 \text{ kg/m}^3 \\ \mu_l &= 1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.1590 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 1448 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.0808 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{\text{sat}} - T_s) \\ &= 168.2 \times 10^3 \text{ J/kg} + 0.68 \times 1448 \text{ J/kg}\cdot^\circ\text{C}(35 - 25)^\circ\text{C} = 178.0 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(1188 \text{ kg/m}^3)(1188 - 43.41 \text{ kg/m}^3)(178.0 \times 10^3 \text{ J/kg})(0.0808 \text{ W/m}\cdot^\circ\text{C})^3}{(1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s})(35 - 25)^\circ\text{C}(0.015 \text{ m})} \right]^{1/4} \\ &= 1880 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the pipe is

$$A_s = \pi DL = \pi(0.015 \text{ m})(7 \text{ m}) = 0.3299 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (1880 \text{ W/m}^2\cdot^\circ\text{C})(0.3299 \text{ m}^2)(35 - 25)^\circ\text{C} = 6200 \text{ W}$$

The rate of condensation of steam is determined from

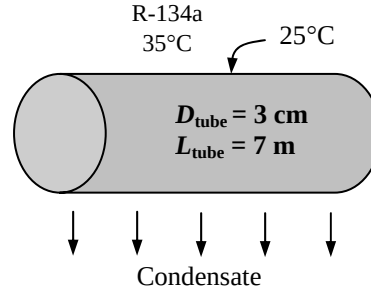
$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{6200 \text{ J/s}}{178.0 \times 10^3 \text{ J/kg}} = 0.0348 \text{ kg/s} = \mathbf{2.09 \text{ kg/min}}$$

10-80 Saturated refrigerant-134a vapor condenses on the outside of a horizontal tube maintained at a specified temperature. The rate of condensation of the refrigerant is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of refrigerant-134a at the saturation temperature of 35°C are $h_{fg} = 168.2 \times 10^3$ J/kg and $\rho_v = 43.41$ kg/m³. The properties of liquid R-134a at the film temperature of $T_f = (T_{sat} + T_s) / 2 = (35 + 25) / 2 = 30^\circ\text{C}$ are (Table A-10),

$$\begin{aligned}\rho_l &= 1188 \text{ kg/m}^3 \\ \mu_l &= 1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.1590 \times 10^{-6} \text{ m}^2/\text{s} \\ C_{pl} &= 1448 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.0808 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68C_{pl}(T_{sat} - T_s) \\ &= 168.2 \times 10^3 \text{ J/kg} + 0.68 \times 1448 \text{ J/kg}\cdot^\circ\text{C}(35 - 25)^\circ\text{C} = 178.0 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{sat} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(1188 \text{ kg/m}^3)(1188 - 43.41 \text{ kg/m}^3)(178.0 \times 10^3 \text{ J/kg})(0.0808 \text{ W/m}\cdot^\circ\text{C})^3}{(1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s})(35 - 25)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 1581 \text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the pipe is

$$A_s = \pi DL = \pi(0.03 \text{ m})(7 \text{ m}) = 0.6597 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{sat} - T_s) = (1581 \text{ W/m}^2\cdot^\circ\text{C})(0.6597 \text{ m}^2)(35 - 25)^\circ\text{C} = 10,428 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{10,428 \text{ J/s}}{178.0 \times 10^3 \text{ J/kg}} = 0.05858 \text{ kg/s} = \mathbf{3.52 \text{ kg/min}}$$

10-81 Saturated steam at 270 kPa pressure and thus at a saturation temperature of $T_{\text{sat}} = 130^\circ\text{C}$ (Table A-9) condenses inside a horizontal tube which is maintained at 110°C . The average heat transfer coefficient and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vapor velocity is low so that $\text{Re}_{\text{vapor}} < 35,000$.

Properties The properties of water at the saturation temperature of 130°C are $h_{fg} = 2174 \times 10^3 \text{ J/kg}$ and $\rho_v = 1.50 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (130 + 110)/2 = 120^\circ\text{C}$ are (Table A-9),

$$\rho_l = 943.4 \text{ kg/m}^3$$

$$\mu_l = 0.232 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.246 \times 10^{-6} \text{ m}^2/\text{s}$$

$$C_{pl} = 4244 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.683 \text{ W/m}\cdot^\circ\text{C}$$

Analysis The condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{internal}} = 0.555 \left[\frac{g\rho_l(\rho_l - \rho_v)k_l^3}{\mu_l(T_{\text{sat}} - T_s)} \left(h_{fg} + \frac{3}{8}C_{pl}(T_{\text{sat}} - T_s) \right) \right]^{1/4} \\ &= 0.555 \left[\frac{(9.8 \text{ m/s}^2)(943.4 \text{ kg/m}^3)(943.4 - 1.50) \text{ kg/m}^3(0.683 \text{ W/m}\cdot^\circ\text{C})^3}{(0.232 \times 10^{-3} \text{ kg/m}\cdot\text{s})(130 - 110)^\circ\text{C}} \right. \\ &\quad \left. \times \left(2174 \times 10^3 \text{ J/kg} + \frac{3}{8}(4244 \text{ kJ/kg}\cdot^\circ\text{C})(130 - 110)^\circ\text{C} \right) \right]^{1/4} \\ &= 3345 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The heat transfer surface area of the pipe is

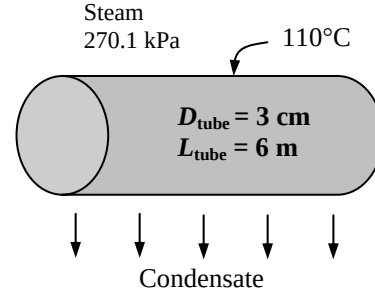
$$A_s = \pi DL = \pi(0.03 \text{ m})(6 \text{ m}) = 0.5655 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (3345 \text{ W/m}^2\cdot^\circ\text{C})(0.5655 \text{ m}^2)(130 - 110)^\circ\text{C} = 37,831 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}} = \frac{37,831 \text{ J/s}}{2174 \times 10^3 \text{ J/kg}} = 0.0174 \text{ kg/s}$$

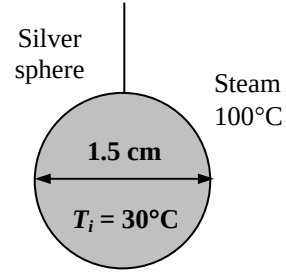


10-82 Saturated steam condenses on a suspended silver sphere which is initially at 30°C. The time needed for the temperature of the sphere to rise to 50°C and the amount of steam condenses are to be determined.

Assumptions 1 The temperature of the sphere changes uniformly and thus the lumped system analysis is applicable. 2 The average condensation heat transfer coefficient evaluated for the average temperature can be used for the entire process. 3 Constant properties at room temperature can be used for the silver ball.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of the silver ball at room temperature and the properties of liquid water at the average film temperature of $T_f = (T_{\text{sat}} + T_{s,\text{ave}}) / 2 = (100 + 40) / 2 = 70^\circ\text{C}$ are (Tables A-3 and A-9),

Silver Ball:	Liquid Water:
$\rho = 10,500 \text{ kg/m}^3$	$\rho_l = 977.5 \text{ kg/m}^3$
$\alpha = 174 \times 10^{-6} \text{ m}^2/\text{s}$	$\mu_l = 0.404 \times 10^{-3} \text{ kg/m}\cdot\text{s}$
$C_p = 235 \text{ J/kg}\cdot^\circ\text{C}$	$C_{pl} = 4190 \text{ J/kg}\cdot^\circ\text{C}$
$k_l = 429 \text{ W/m}\cdot^\circ\text{C}$	$k_l = 0.663 \text{ W/m}\cdot^\circ\text{C}$



Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4190 \text{ J/kg}\cdot^\circ\text{C} (100 - 40)^\circ\text{C} = 2428 \times 10^3 \text{ J/kg} \end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{sph}} = 0.813 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.813 \left[\frac{(9.8 \text{ m/s}^2)(977.5 \text{ kg/m}^3)(977.5 - 0.60 \text{ kg/m}^3)(2428 \times 10^3 \text{ J/kg})(0.663 \text{ W/m}\cdot^\circ\text{C})^3}{(0.404 \times 10^{-3} \text{ kg/m}\cdot\text{s})(100 - 40)^\circ\text{C}(0.015 \text{ m})} \right]^{1/4} \\ &= 9445 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The characteristic length and the Biot number for the lumped system analysis is (see Chap. 4)

$$\begin{aligned} L_c &= \frac{V}{A} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{0.015 \text{ m}}{6} = 0.0025 \text{ m} \\ Bi &= \frac{h L_c}{k} = \frac{(9445 \text{ W/m}^2\cdot^\circ\text{C})(0.0025 \text{ m})}{(429 \text{ W/m}\cdot^\circ\text{C})} = 0.055 < 0.1 \end{aligned}$$

The lumped system analysis is applicable since $Bi < 0.1$. Then the time needed for the temperature of the sphere to rise from 30 to 50°C is determined to be

$$\begin{aligned} b &= \frac{h A_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{9445 \text{ W/m}^2\cdot^\circ\text{C}}{(10,500 \text{ kg/m}^3)(235 \text{ J/kg}\cdot^\circ\text{C})(0.0025 \text{ m})} = 1.531 \text{ s}^{-1} \\ \frac{T(t) - T_\infty}{T_i - T_\infty} &= e^{-bt} \longrightarrow \frac{50 - 100}{30 - 100} = e^{-1.531t} \longrightarrow t = \mathbf{0.22 \text{ s}} \end{aligned}$$

The total heat transfer to the ball and the amount of steam that condenses become

$$\begin{aligned} m_{\text{sphere}} &= \rho V = \rho \frac{\pi D^3}{6} = (10,500 \text{ kg/m}^3) \frac{\pi (0.015 \text{ m})^3}{6} = 0.0186 \text{ kg} \\ Q &= m C_p [T(t) - T_i]_{\text{sphere}} = (0.0186 \text{ kg})(235 \text{ J/kg}\cdot^\circ\text{C})(50 - 30)^\circ\text{C} = 87.2 \text{ J} \\ \dot{m}_{\text{condensation}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{87.2 \text{ J/s}}{2428 \times 10^3 \text{ J/kg}} = \mathbf{0.0359 \times 10^{-3} \text{ kg/s}} \end{aligned}$$

10-83 Steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a suspended silver sphere which is initially at 30°C . The time needed for the temperature of the sphere to rise to 50°C and the amount of steam condenses during this process are to be determined.

Assumptions 1 The temperature of the sphere changes uniformly and thus the lumped system analysis is applicable. 2 The average condensation heat transfer coefficient evaluated for the average temperature can be used for the entire process. 3 Constant properties at room temperature can be used for the silver ball.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of the silver ball at room temperature and the properties of liquid water at the average film temperature of $T_f = (T_{\text{sat}} + T_{s,\text{ave}})/2 = (100 + 40)/2 = 70^\circ\text{C}$ are (Tables A-3 and A-9),

Silver Ball:

$$\rho = 10,500 \text{ kg/m}^3$$

$$\alpha = 174 \times 10^{-6} \text{ m}^2/\text{s}$$

$$C_p = 235 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 429 \text{ W/m}\cdot^\circ\text{C}$$

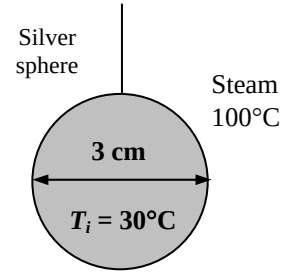
Liquid Water:

$$\rho_l = 977.5 \text{ kg/m}^3$$

$$\mu_l = 0.404 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

$$C_{pl} = 4190 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.663 \text{ W/m}\cdot^\circ\text{C}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 C_{pl} (T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4190 \text{ J/kg}\cdot^\circ\text{C} (100 - 40)^\circ\text{C} = 2428 \times 10^3 \text{ J/kg} \end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{sph}} = 0.813 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.813 \left[\frac{(9.8 \text{ m/s}^2)(977.5 \text{ kg/m}^3)(977.5 - 0.60 \text{ kg/m}^3)(2428 \times 10^3 \text{ J/kg})(0.663 \text{ W/m}\cdot^\circ\text{C})^3}{(0.404 \times 10^{-3} \text{ kg/m}\cdot\text{s})(100 - 40)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 7942 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The characteristic length and the Biot number for the lumped system analysis is (see Chap. 4)

$$\begin{aligned} L_c &= \frac{V}{A_s} = \frac{\pi D^3 / 6}{\pi D^2} = \frac{D}{6} = \frac{0.03 \text{ m}}{6} = 0.005 \text{ m} \\ Bi &= \frac{h L_c}{k} = \frac{(7942 \text{ W/m}^2\cdot^\circ\text{C})(0.005 \text{ m})}{(429 \text{ W/m}\cdot^\circ\text{C})} = 0.093 < 0.1 \end{aligned}$$

The lumped system analysis is applicable since $Bi < 0.1$. Then the time needed for the temperature of the sphere to rise from 30 to 50°C is determined to be

$$\begin{aligned} b &= \frac{h A_s}{\rho C_p V} = \frac{h}{\rho C_p L_c} = \frac{7942 \text{ W/m}^2\cdot^\circ\text{C}}{(10,500 \text{ kg/m}^3)(235 \text{ J/kg}\cdot^\circ\text{C})(0.005 \text{ m})} = 0.644 \text{ s}^{-1} \\ \frac{T(t) - T_\infty}{T_i - T_\infty} &= e^{-bt} \longrightarrow \frac{50 - 100}{30 - 100} = e^{-0.644t} \longrightarrow t = \mathbf{0.52 \text{ s}} \end{aligned}$$

The total heat transfer to the ball and the amount of steam that condenses become

$$\begin{aligned} m_{\text{sphere}} &= \rho V = \rho \frac{\pi D^3}{6} = (10,500 \text{ kg/m}^3) \frac{\pi (0.03 \text{ m})^3}{6} = 0.148 \text{ kg} \\ Q &= m C_p [T(t) - T_i]_{\text{sphere}} = (0.148 \text{ kg})(235 \text{ J/kg}\cdot^\circ\text{C})(50 - 30)^\circ\text{C} = 698 \text{ J} \\ \dot{m}_{\text{condensation}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{698 \text{ J/s}}{2428 \times 10^3 \text{ J/kg}} = \mathbf{0.287 \times 10^{-3} \text{ kg/s}} \end{aligned}$$

10-84 Saturated steam at a saturation temperature of $T_{\text{sat}} = 95^\circ\text{C}$ (Table A-9) condenses on a canned drink at 5°C in a dropwise manner. The heat transfer coefficient for this dropwise condensation is to be determined.

Assumptions The heat transfer coefficient relation for dropwise condensation that was developed for copper surfaces is also applicable for aluminum surfaces.

Analysis Noting that the saturation temperature is less than 100°C , the heat transfer coefficient for dropwise condensation can be determined from Griffith's relation to be

$$h = h_{\text{dropwise}} = 51,104 + 2044T_{\text{sat}} = 51,104 + 2044 \times 95 = \mathbf{245,284 \text{ W/m}^2 \cdot ^\circ\text{C}}$$



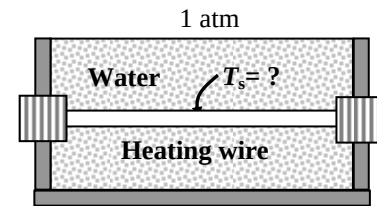
10-85 Water is boiled at 1 atm pressure and thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a nickel electric heater whose diameter is 2 mm. The highest temperature at which this heater can operate without burnout is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the water are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned} \rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0589 \text{ N/m} & C_{pl} &= 4217 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.75 \end{aligned}$$

Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel surface (Table 10-3).



Analysis The maximum rate of heat transfer without the burnout is simply the critical heat flux. For a horizontal heating wire, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned} L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.001) \left(\frac{9.8(957.9 - 0.60)}{0.0589} \right)^{1/2} = 0.399 < 1.2 \\ C_{cr} &= 0.12L^{*-0.25} = 0.12(0.399)^{-0.25} = 0.151 \end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned} \dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} = 0.151 (2257 \times 10^3) [0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= \mathbf{1,280,000 \text{ W/m}^2} \end{aligned}$$

Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into Rohsenow relation together with other properties gives

$$\begin{aligned} \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{C_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,280,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.8(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0060(2257 \times 10^3)1.75} \right)^3 \end{aligned}$$

It gives the maximum temperature to be: $T_s = \mathbf{109.6^\circ\text{C}}$

